
SVD-Free Tensor Completion: Surrogate Gaps and Cyclic Optimization

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Abstract

Tensor completion recovers missing entries in multiway data using low-rank structure. We study a smooth penalty on CANDECOMP/PARAFAC (CP) factors that avoids singular value decompositions. The penalty generally upper-bounds tensor nuclear norms rather than representing them exactly. We explain this gap through decomposition choice and component scaling, characterize balanced factors, and show that width changes regularization even for a fixed rank-one tensor. Strongly convex quadratic factor subproblems enable exact, gradient, and hybrid cyclic updates. At fixed finite width and positive regularization, these methods converge to a stationary point with an $O(1/K)$ bound on the minimum squared gradient norm. Reproducible synthetic experiments test the gap formulas, width dependence, and optimizer progress; historical real-data comparisons provide supplementary evidence. The analysis distinguishes optimization cost, norm approximation, and reconstruction accuracy.

1 INTRODUCTION

Tensor completion reconstructs missing entries in multiway data such as hyperspectral images, videos, and traffic measurements. For example, a video can be arranged as a tensor indexed by height, width, and time. Low-rank models share latent structure across these modes, allowing observations at some coordinates to inform predictions at others (Kolda and Bader, 2009; Liu et al., 2012). The challenge is to

regularize this structure without making large-scale optimization prohibitively costly. Matrix completion offers a useful starting point: a quadratic penalty on two low-rank factors can exactly represent the nuclear norm when the factor width is sufficient (Recht, 2011). Extending this construction to a shared CANDECOMP/PARAFAC (CP) model avoids large singular value decompositions (SVDs). For tensors, however, the analogous factor penalty generally exceeds both the atomic and average unfolding nuclear norms. Cheap optimization therefore need not mean that the intended tensor norm is represented exactly.

We ask three questions. What causes this excess penalty? How does the number of available CP components affect it? Can the resulting completion objective still be optimized efficiently with a convergence guarantee? Our contributions are threefold.

- We separate the surrogate-atomic norm gap into two nonnegative terms: excess component weight from the chosen decomposition and excess penalty from component scaling. We characterize the optimal product-preserving scaling and prove that unconstrained stationary points have balanced factor norms, while explaining why balancing alone does not make the surrogate exact. For the average unfolding nuclear norm, we express the gap through Bures–Wasserstein distances between factor Gram matrices.
- We show that factor width controls both representation capacity and the least attainable regularization penalty. Increasing width can lower the penalty even when the represented tensor remains unchanged and has rank one. We characterize this rank-one penalty through a scalar allocation problem and distinguish nesting of global minima from the behavior of locally optimized models.
- We derive SVD-free updates by exploiting strongly convex quadratic factor subproblems. For fixed finite width and positive regularization, we prove convergence to a stationary point

and an $\mathcal{O}(1/K)$ bound on the minimum squared gradient norm. Controlled experiments test the gap identities, width effects, and finite-budget optimizer progress, measuring objective reduction, norm agreement, and held-out reconstruction error separately.

Section 2 introduces the required notation and matrix analogy. Sections 4–5 develop the surrogate, its gaps and width dependence, and cyclic optimization. Section 6 evaluates these distinct claims; the supplement contains full proofs and protocols.

2 BACKGROUND AND NOTATION

Tensors and CP factors. An order- N tensor is an array with N modes. For $\mathbf{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$, let Ω denote the observed coordinates. The operator P_Ω keeps those entries and sets the others to zero, so a masked squared loss uses only observations. A rank-one tensor is an outer product: for three modes, $(\mathbf{a} \circ \mathbf{b} \circ \mathbf{c})_{ijk} = a_i b_j c_k$. A CP model adds R such components (Kolda and Bader, 2009):

$$\mathcal{X}(\mathcal{A}) = \sum_{r=1}^R \mathbf{a}_{1r} \circ \dots \circ \mathbf{a}_{Nr}, \quad \mathcal{A} = (\mathbf{A}_1, \dots, \mathbf{A}_N). \quad (1)$$

Here $\mathbf{A}_i = [\mathbf{a}_{i1}, \dots, \mathbf{a}_{iR}] \in \mathbb{R}^{I_i \times R}$. The *width* R is the number of available columns; the CP rank is the smallest number needed to represent a tensor. They need not agree.

Unfoldings and nuclear norms. The mode- i unfolding $\mathbf{X}_{(i)}$ is an $I_i \times \prod_{j \neq i} I_j$ matrix: mode i indexes its rows and the remaining coordinates index its columns. Write $\mathbf{B}_i = \odot_{j \neq i} \mathbf{A}_j$ for the Khatri–Rao product, whose r th column is the Kronecker product of the other r th factor columns. Then $\mathbf{X}_{(i)} = \mathbf{A}_i \mathbf{B}_i^\top$. The matrix nuclear norm $\|\mathbf{X}\|_*$ is the sum of singular values; $\|\cdot\|_F$ is the Frobenius norm. For tensors, we distinguish *atomic* and *average unfolding* norms:

$$\|\mathcal{X}\|_{\text{at}} = \inf_{\mathbf{X} = \sum_r \alpha_r \mathbf{u}_r} \sum_r |\alpha_r|, \quad \|\mathcal{X}\|_{\text{unf}} = \frac{1}{N} \sum_i \|\mathbf{X}_{(i)}\|_*. \quad (2)$$

The infimum ranges over all finite decompositions into unit rank-one atoms $\mathbf{u}_r = \mathbf{u}_{1r} \circ \dots \circ \mathbf{u}_{Nr}$ with $\|\mathbf{u}_{ir}\|_2 = 1$. The atomic norm measures the least total component weight; the unfolding norm averages matrix penalties. They agree for matrices, but generally differ for $N > 2$.

Why use a factor surrogate? For a matrix, the identity $\|\mathbf{X}\|_* = \min_{\mathbf{A}\mathbf{B}^\top = \mathbf{X}} (\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2)/2$ holds at

sufficient width (Recht, 2011). Applying its inequality to each CP unfolding gives an inexpensive upper-bound penalty. Equality is not automatic: the factors of all unfoldings must come from the same CP components. The next section quantifies this discrepancy.

3 RELATED WORK

Tensor norms and factor regularization. Unfolding nuclear-norm methods, including HaLRTC (Liu et al., 2012), provide convex completion formulations; t-SVD completion uses a distinct tubal-rank model (Zhang and Aeron, 2017). The atomic nuclear norm instead retains rank-one tensor atoms (Friedland and Lim, 2018), following the structured atomic-gauge framework of Chandrasekaran et al. (2012). Tensor spectral-norm computation is NP-hard (Hillar and Lim, 2013), motivating tractable bounds and approximate spectral oracles. Relations between atomic, unfolding, and Frobenius norms are established by Hu (2015); in particular, the fiber upper bound used in our Gaussian analysis is known. Trace-norm regularized CP penalizes nuclear norms of the smaller factor matrices (Liu et al., 2015), reducing the dimensions of its spectral subproblems.

Variational penalties and gap geometry. Exact factor variational forms are also known. Cheng et al. (2016) represent the atomic norm using sums of N th powers of Euclidean factor-vector norms, minimized over sufficiently wide CP factorizations, and optimize with approximate spectral oracles inside generalized conditional gradients. Lacroix et al. (2018) establish norm balancing for homogeneous factor penalties and distinguish squared-Frobenius CP regularization from tensor nuclear norms. Fan et al. (2023) develop the connection between Euclidean factor penalties and tensor Schatten- p quasi-norms, including recovery-error analysis. These results already provide SVD-free factor regularization and balancing arguments. Our balanced component cost instead combines the amplitude powers $2/N$ and $2 - 2/N$ within the same decomposition. We characterize the resulting mixed penalty’s atomic and unfolding gaps and width dependence while retaining strongly convex quadratic factor blocks. Our unfolding-gap formula applies a classical matrix identity (Bhatia et al., 2019) to CP factors; that identity is not new.

Scalable completion methods. CP-WOPT optimizes observed-entry CP least squares (Acar et al., 2011); TMac uses separate unfolding factorizations (Xu et al., 2015). Other approaches include Bayesian CP (Rai et al., 2014), Tucker Riemannian and scaled-gradient methods (Kasai and Mishra, 2016; Tong et al.,

2022), approximate Richardson iterations (Ghadiri et al., 2025), and neural completion (Liu et al., 2019). These use different model classes or assumptions; our claims concern the specified CP surrogate.

Block optimization and guarantees. Our convergence analysis follows established descent and KL arguments (Attouch et al., 2013; Bolte et al., 2014). We verify their requirements for this polynomial objective and its mixed exact and gradient updates. The resulting guarantee holds for each fixed R without requiring correct rank specification. It is a stationarity result: it does not imply global optimality, or a rate constant uniform over widths. This scope differs from local recovery results under sampling and initialization assumptions (Tong et al., 2022).

4 FACTOR SURROGATE & GAPS

Using the notation of Section 2, fix $N \geq 2$, a finite width R , and $\lambda > 0$. Set $q_{ir} = \|\mathbf{a}_{ir}\|_2^2$. Averaging the matrix factor penalty across CP unfoldings gives

$$\mathcal{R}(\mathcal{A}) = \frac{1}{2N} \sum_{i=1}^N \sum_{r=1}^R \left(q_{ir} + \prod_{j \neq i} q_{jr} \right). \quad (3)$$

We minimize the observed-entry completion objective

$$F(\mathcal{A}) = \frac{1}{2} \|P_\Omega(\mathcal{X}(\mathcal{A}) - \mathcal{Y})\|_F^2 + \lambda \mathcal{R}(\mathcal{A}). \quad (4)$$

The identity $\mathcal{R} = (2N)^{-1} \sum_i (\|\mathbf{A}_i\|_F^2 + \|\mathbf{B}_i\|_F^2)$ explains the construction. Its evaluation needs only column norms, not explicit unfoldings, Khatri–Rao matrices, or SVDs.

4.1 Exact decomposition and optimal scaling

Define $u_{ir} = \|\mathbf{a}_{ir}\|_2$ and $\mu_r = \prod_i u_{ir}$. Normalizing nonzero components gives an atomic decomposition. Therefore,

$$\max_i \|\mathbf{X}_{(i)}\|_* \leq \|\mathcal{X}\|_{\text{at}} \leq \sum_r \mu_r \leq \mathcal{R}. \quad (5)$$

A *gap* is the excess of $\mathcal{R}$ over the reference norm for the same tensor. Decomposition choice can add excess component weight; rescaling factor vectors can add further penalty without changing the tensor. The following result separates these effects.

Proposition 1 (Atomic gap and balanced scaling). *For any CP factorization, including those with zero components, let $u_{ir} = \|\mathbf{a}_{ir}\|_2$ and $\mu_r = \prod_i u_{ir}$. The*

surrogate-atomic norm gap decomposes as

$$\mathcal{R} - \|\mathcal{X}\|_{\text{at}} = \underbrace{\sum_r \mu_r}_{\Delta_{\text{dec}}} - \|\mathcal{X}\|_{\text{at}} + \Delta_{\text{scale}}, \quad (6)$$

$$\Delta_{\text{scale}} = \frac{1}{2N} \sum_{r,i} \left(u_{ir} - \prod_{j \neq i} u_{jr} \right)^2. \quad (7)$$

Both terms are nonnegative: Δ_{dec} measures excess component weight relative to the atomic norm, while Δ_{scale} measures the factor penalty above that weight.

For a nonzero component of magnitude μ , positive rescalings whose multipliers have product one leave the component unchanged. Among these rescalings, its penalty is minimized when $u_1 = \dots = u_N = \mu^{1/N}$. The minimum penalty and its excess over μ are

$$\phi_N(\mu) = \frac{1}{2} (\mu^{2/N} + \mu^{2-2/N}), \quad (8)$$

$$g_N(\mu) := \phi_N(\mu) - \mu = \frac{1}{2} (\mu^{1/N} - \mu^{1-1/N})^2, \quad (9)$$

respectively. Thus, balancing minimizes the scaling gap but need not eliminate it.

For $\lambda > 0$, every unconstrained stationary point of (4) has balanced component norms $u_{ir} = \mu_r^{1/N}$. Each component is therefore either nonzero in every mode or entirely zero; for the latter, we set $\phi_N(0) = g_N(0) = 0$.

The decomposition gap measures whether the chosen CP components attain the least atomic weight. The scaling gap measures the cost of their factor norms and need not vanish even after balancing. The first identity follows by completing squares. The scaling minimum follows from arithmetic–geometric mean inequalities. To see stationary balancing, multiply two vectors of one component by e^t and e^{-t} . This preserves $\mathcal{X}$, and the derivative of F at zero is

$$\frac{\lambda}{N} (q_{ir} - q_{jr}) \left(1 + \prod_{\ell \notin \{i,j\}} q_{\ell r} \right). \quad (10)$$

The positive bracket forces equal norms. Complete proofs, including zero cases, are in the supplement.

For $N = 2$, $g_2 \equiv 0$ and sufficient width recovers matrix nuclear norm form. For $N > 2$, $g_N(\mu) = 0$ only at $\mu = 1$ when $\mu > 0$. Thus balancing removes avoidable scaling imbalance but does not generally make the surrogate exact. Computable interval

$$\mathcal{R} - \sum_r \mu_r \leq \mathcal{R} - \|\mathcal{X}\|_{\text{at}} \leq \mathcal{R} - \max_i \|\mathbf{X}_{(i)}\|_* \quad (11)$$

can assess factors without computing atomic norm.

Unfolding gap. Let $\mathbf{P}_i = \mathbf{A}_i^\top \mathbf{A}_i$ and $\mathbf{Q}_i = \mathbf{B}_i^\top \mathbf{B}_i$. These Gram matrices record inner products between component columns. Thus, the unfolding gap depends on component correlations as well as individual norms. Using the Bures–Wasserstein metric (Bhatia et al., 2019), the matrix nuclear norm identity gives

$$\mathcal{R} - \|\mathcal{X}\|_{\text{unf}} = \frac{1}{2N} \sum_i d_{\text{BW}}^2(\mathbf{P}_i, \mathbf{Q}_i), \quad (12)$$

where $d_{\text{BW}}^2(P, Q) = \text{tr } P + \text{tr } Q - 2\|P^{1/2}Q^{1/2}\|_*$. This holds for singular positive semidefinite Gram matrices; the gap vanishes if and only if $\mathbf{P}_i = \mathbf{Q}_i$ for every mode.

Gaussian reference scales. Let $\mathcal{G} \in \mathbb{R}^{n \times \dots \times n}$ have independent entries distributed as $\mathcal{N}(0, \sigma^2)$. For fixed order $N \geq 2$, as $n \rightarrow \infty$, our norm comparison gives

$$\mathbb{E}\|\mathcal{G}\|_{\text{at}} = \Theta_N(\sigma n^{N-1/2}), \mathbb{E}\|\mathcal{G}\|_{\text{unf}} = \Theta_N(\sigma n^{(N+1)/2}). \quad (13)$$

Their expectation ratio scales as $n^{(N-2)/2}$, illustrating why the norms differ beyond matrices (where $N = 2$). Appendix A.4 gives finite-dimensional certificates, including the atomic upper bound U_δ , and Section 6.1 tests them. This comparison concerns tensor norms, not the surrogate gap at learned factors. **The expected scaling-gap formula for lognormal component magnitudes** is also given in the supplement.

4.2 Role of the factor width

The parameter R counts available CP columns; it bounds, but need not equal, the represented tensor’s CP rank. Let

$$\mathcal{P}_R(\mathcal{X}) := \min_{[\mathbf{A}_1, \dots, \mathbf{A}_N] = \mathcal{X}} \mathcal{R}(\mathcal{A}),$$

with value $+\infty$ when no width- R representation exists. Also let J_R^* be the global minimum of (4) at width R and fixed $\lambda > 0$, and use ϕ_N from (8).

Proposition 2 (Width dependence). *For $N \geq 2$, J_R^* is attained and $J_{R+1}^* \leq J_R^*$. Whenever $\mathcal{P}_R(\mathcal{X})$ is finite, its minimum is attained, and $\mathcal{P}_{R+1}(\mathcal{X}) \leq \mathcal{P}_R(\mathcal{X})$. For a unit rank-one atom $\mathcal{U}$ and $\mu \geq 0$,*

$$\mathcal{P}_R(\mu \mathcal{U}) = \min_{\substack{\nu_r \geq 0 \\ \sum_{r=1}^R \nu_r = \mu}} \sum_{r=1}^R \phi_N(\nu_r). \quad (14)$$

Consequently, for $N > 2$ and $\mu > 0$,

$$\mathcal{P}_R(\mu \mathcal{U}) = \mu \iff \mu \in \{1, \dots, R\}.$$

The proof is in Appendix G. Thus, width can change the effective tensor penalty even when the tensor’s CP rank remains one. For $N > 2$, for example,

Algorithm 1 Cyclic Block Optimization (CBO)

Input: Observations $\{(\omega, y_\omega) : \omega \in \Omega\}$, width R , $\lambda > 0$

1. Initialize $\mathcal{A}^0$ and its column norms; set $k \leftarrow 0$.
 2. **While** the stopping criterion is not satisfied:
 - (a) **For** $i = 1, \dots, N$, in order:
 - Compute $\mathbf{D}_i$ from the column norms q_{jr} .
 - Accumulate $\mathbf{G}_{i,s}, \mathbf{c}_{i,s}$ by (15) and solve (16) by Cholesky factorization.
 - Refresh the column norms of $\mathbf{A}_i$.
 - (b) Set $\mathcal{A}^{k+1}$ to the updated factors; set $k \leftarrow k + 1$.
 3. **Return** $\hat{\mathcal{X}} = [\mathbf{A}_1^k, \dots, \mathbf{A}_N^k]$.
-

$\mathcal{P}_1(2\mathcal{U}) = \phi_N(2) > 2$, whereas $\mathcal{P}_R(2\mathcal{U}) = 2$ whenever $R \geq 2$. In contrast, for $0 < \mu \leq 1$, $\mathcal{P}_R(\mu \mathcal{U}) = \phi_N(\mu)$ for every width. The appendix also analyzes equal-split constructions, without assuming that they globally minimize (14).

5 CYCLIC OPTIMIZATION

A block is one entire factor matrix $\mathbf{A}_i$. A sweep updates all N blocks in order, holding the others fixed at their latest values. Although the joint objective is nonconvex, each block problem has a unique minimizer because of its positive quadratic regularization.

Fixing all factors except $\mathbf{A}_i$ makes (4) a strongly convex quadratic. For an observed coordinate ω , define the elementwise product $\mathbf{b}_\omega^{(i)} = \odot_{j \neq i} \mathbf{A}_j[\omega_j, :]^\top$. Define the diagonal matrix $\mathbf{D}_i$ by $d_{ir} = \sum_{j \neq i} \prod_{\ell \notin \{i,j\}} q_{\ell r}$ and set

$$\mathbf{G}_{is} = \sum_{\omega \in \Omega: \omega_i = s} \mathbf{b}_\omega^{(i)} (\mathbf{b}_\omega^{(i)})^\top, \mathbf{c}_{is} = \sum_{\omega \in \Omega: \omega_i = s} y_\omega \mathbf{b}_\omega^{(i)}. \quad (15)$$

An exact block update solves independent systems,

$$\left[\mathbf{G}_{is} + \frac{\lambda}{N} (\mathbf{I}_R + \mathbf{D}_i) \right] \mathbf{A}_i[s, :]^\top = \mathbf{c}_{is}. \quad (16)$$

A row with no observations is set to zero. Algorithm 1 summarizes the exact variant; the gradient alternative follows below.

Alternatively, use $\mathbf{A}_i^+ = \mathbf{A}_i - L_i^{-1} \nabla_i F$ with

$$L_i = \sum_r \prod_{j \neq i} q_{jr} + \frac{\lambda}{N} (1 + \max_r d_{ir}), \quad (17)$$

a valid upper bound on the block Hessian norm. Products are evaluated without division, so zero columns are permitted. Cyclic updates always use the latest preceding blocks. We call the all-exact variant *Cyclic Block Optimization* (CBO); exact and gradient blocks may also be combined.

Theorem 1 (Stationary convergence). *For $N \geq 2$, fixed finite R , and $\lambda > 0$, cyclic exact updates, gradient updates using (17), and their hybrids decrease F , have finite length, and converge to a stationary point. More generally, gradient updates may use any valid block smoothness bounds that remain uniformly bounded along the iterates. If $E_0 = F(\mathcal{A}^0) - \inf F$, $\mu = \lambda/N$, and Λ bounds the full-gradient Lipschitz constant on convex hull of the initial sublevel set, then*

$$\min_{0 \leq k < K} \|\nabla F(\mathcal{A}^{k+1})\|^2 \leq \frac{4(\bar{L}^2 + N\Lambda^2)E_0}{\mu K}, \quad (18)$$

where $\bar{L}$ bounds the gradient-step smoothness parameters (any positive value suffices for all-exact updates).

The key facts are block curvature at least λ/N , coercivity $F \geq \lambda\|\mathcal{A}\|^2/(2N)$, and polynomial structure. The latter supplies a bounded Hessian on the compact sublevel hull and the Kurdyka–Łojasiewicz property. Sufficient decrease and a relative gradient bound then satisfy the conditions of Attouch et al. (2013); full details are in the supplement. Stationarity means $\nabla F = 0$; it does not imply global minimization.

Dependence on width. With $M = |\Omega|$, streamed exact sweeps have conservative cost

$$O\left(N^2MR + NMR^2 + N^2R + R^3 \sum_i I_i\right). \quad (19)$$

Gradient sweeps using (17) cost $O(N^2MR + N^2R + R \sum_i I_i)$ and avoid row Hessians. The supplement specifies storage and alternative Gram-based bounds. These bounds motivate measuring accuracy and computational cost together when selecting R .

6 EXPERIMENTS

We evaluate the gap analysis and exact cyclic block optimization (CBO) through synthetic norm checks, controlled completion fits across widths, and real-data comparisons at solver-time targets.

6.1 Gaussian Norm Benchmarks

For orders three and four, we sample tensors with independent $\mathcal{N}(0, 1)$ entries at six equal-mode sizes per order. At each size, 200 draws evaluate unfolding norms and 30 separate draws evaluate atomic certificates, with $\delta = 0.05$. Appendix J gives sizes, seeds, and protocols. Figure 2 shows that all observed unfolding norms lie within their probability interval. The fitted growth exponents are 2.008 and 2.506, consistent with the predicted 2 and $5/2$. The expectation upper bound is 1.0005–1.0183 times the sample mean

across tested sizes. These finite-sample checks complement the analytic bounds. For the atomic norm, the maximum unfolding nuclear norm gives a lower certificate, while the smallest sum of slice nuclear norms over mode pairs gives a feasible-decomposition upper certificate. The revised Gaussian upper bound U_δ is 1.191–1.428 times the mean slice certificate; every sampled slice certificate lies below it. Compared with the coordinate bound, U_δ improves by factors 1.649–5.538.

6.2 Gap Identities & Amplitude Dependence

Figure 3 tests 400 order-three/four CP factorizations with Gaussian directions and lognormal scales, including zero components. The maximum scalar-identity discrepancy normalized by $\mathcal{R}$ is 8.46×10^{-16} . The Bures-identity discrepancies are at most 5.78×10^{-16} in the full-rank group and 2.35×10^{-9} in the singular group, classified from the numerical column ranks of the factors. Singular Gram square roots exhibit greater floating-point sensitivity. For orders three through five and 17 amplitudes from 10^{-2} to 10^2 , numerical scaling minimization agrees with $g_N(\mu)/\mu$ to within 3.55×10^{-15} . For a single normalized rank-one component, the atomic norm equals μ , so this also evaluates its optimized surrogate-atomic gap. Separately, 2×10^6 lognormal samples per setting agree with the exact expected scaling gap within 1.52 Monte Carlo standard errors. This last result concerns the scaling gap; a total-gap interpretation requires atomic norm attainment. Certified gap intervals for the random CP factorizations are reported in the supplement.

6.3 Width, Recovery, and Fitted Gaps

We generate 16^3 tensors of exact CP rank four using orthonormal mode vectors and amplitudes (8, 6, 4, 2). Five paired seeds share tensors, noise, and held-out entries across $R \in \{1, 2, 4, 6, 8, 12\}$ and 20% or 40% training observations. Noise has standard deviation 5% of the signal RMS. A disjoint 10% validation set chooses among three regularization values; each has two nonzero initializations, with the lower training objective selecting the start. Test entries are never used for selection. The 360 fits use exact cyclic updates and a common relative full-gradient tolerance of 10^{-5} , capped at 1,000 sweeps. Only 31 of the 60 selected fits meet that tolerance; all capped fits remain in the reported finite-budget results. Appendix H gives the full protocol and independent implementation checks.

Figure 4 exposes three effects of width. Widths below four cannot represent the true full tensor and have large held-out errors. At $R = 4$, mean test relative errors are 0.033 and 0.018 for the two observation rates; at $R = 12$ they increase to 0.088 and 0.033. Mean-



Figure 1: Qualitative completion of a third-order grayscale video. The left three images show partially observed frames; the right three show their corresponding reconstructions in the same temporal order.

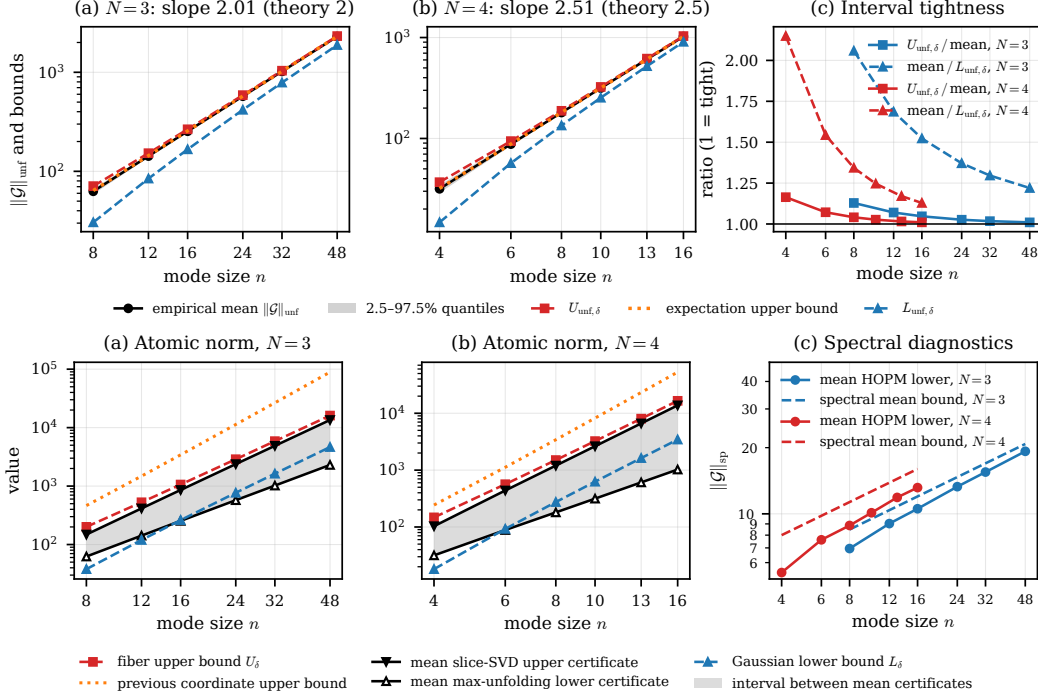


Figure 2: Gaussian norm benchmarks for tensors with independent $\mathcal{N}(0,1)$ entries and equal mode size n , with $\delta = 0.05$. Top (a,b): unfolding-norm sample means and empirical 2.5–97.5% quantiles over 200 draws per size, compared with probability and expectation bounds; top (c): interval tightness ratios. Bottom (a,b): atomic-norm bounds and mean unfolding lower/slice-SVD upper certificates over 30 separate draws per size. Their shaded interval brackets the unknown sample-mean atomic norm; it is not a confidence interval. Bottom (c): mean higher-order power method (HOPM) spectral lower estimates versus an expectation upper bound. The fiber bound improves on the coordinate bound.

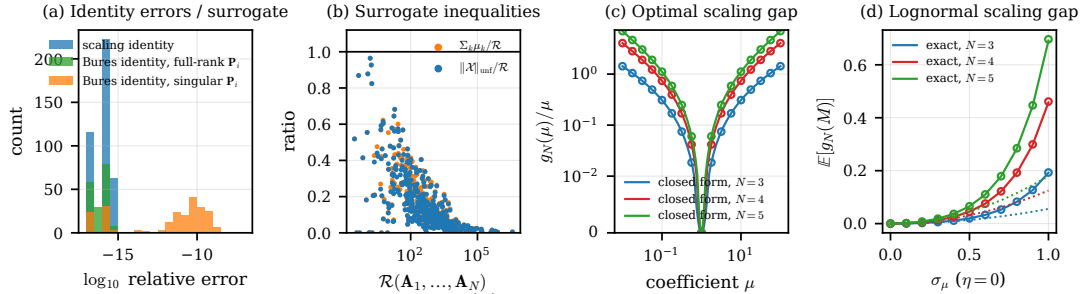


Figure 3: Numerical checks of the gap formulas. (a) Absolute scalar and Bures identity residuals divided by the surrogate, for 400 synthetic CP factorizations; singular Gram matrices increase numerical square-root sensitivity. (b) Component-weight sums and unfolding norms divided by the surrogate, both bounded by one. (c) Minimum relative scaling gap $g_N(\mu)/\mu$: exact curves and numerical minimization results (circles). (d) Expected scaling gap for $\log M \sim \mathcal{N}(0, \sigma_\mu^2)$: exact curves, Monte Carlo estimates (circles), and small-variance approximations (dotted). Panels (c,d) use $N = 3, 4, 5$; (d) concerns scaling rather than the total gap of arbitrary decompositions.

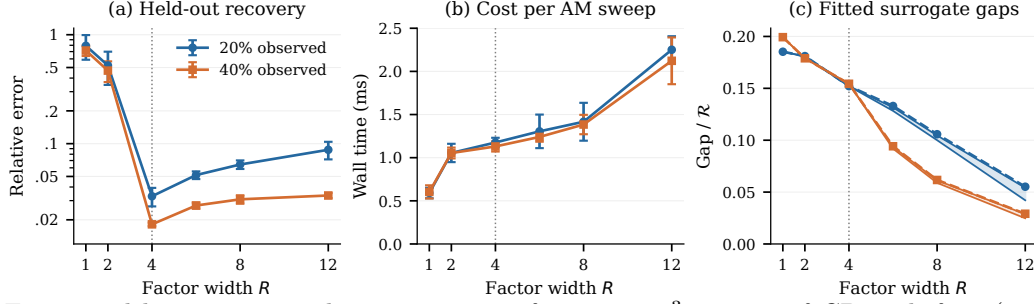


Figure 4: Factor width, recovery, and surrogate gaps for noisy 16^3 tensors of CP rank four (vertical dotted line), with 20% or 40% training observations. (a) Clean-test relative error and (b) time per exact sweep: means and sample standard deviations over five paired seeds. (c) Mean atomic-gap certificate endpoints (shading) and mean exact unfolding gaps (dashed), normalized by each fit’s surrogate; shading is not statistical uncertainty. Recovery is best at $R = 4$; larger widths cost more per sweep yet have smaller relative gaps at different fitted tensors. Validation-selected fits include runs reaching the iteration cap.

while, cost per sweep increases by $1.91\times$ and $1.88\times$. The mean certified atomic-gap intervals and exact unfolding gaps, each normalized by the fitted surrogate, decrease with width despite worse recovery above rank four. These are comparisons of relative gaps at different fitted tensors. Rank-one splitting constructions in the supplement separately demonstrate the width-dependent penalty for an unchanged tensor.

6.4 Real-Tensor Completion

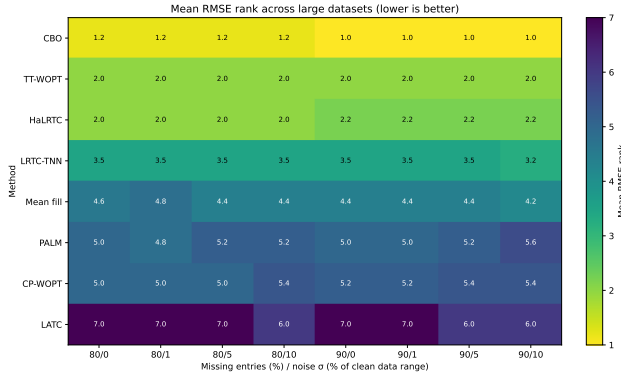


Figure 5: Mean RMSE ranks at the nominal five-second solver target; lower is better. Methods are ranked within each dataset using mean RMSE over five masks, then ranks are averaged over applicable datasets. Columns show missingness/noise standard deviation (% of entries/% of clean data range). Averages cover five datasets for general methods, City only for TT-WOPT, and the two traffic tensors for LATC; these rows therefore use different comparison sets.

The supplied completion suite contains CAVE Balloons, Xiph Bus, Seattle and Portland traffic, and fourth-order Xiph City, spanning 1.82–5.47 million entries. Tensors are normalized to $[0, 1]$. Five shared seeds generate random masks with 80% or 90% miss-

Table 1: Clean 80%-missing RMSE at a nominal five-second target. Means over five shared masks; the supplement gives standard deviations. The comparator is the strongest applicable iterative baseline per row.

Tensor	CBO	Comparator	Method
CAVE	0.03737	0.05306	HaLRTC
Bus	0.15728	0.14022	HaLRTC
Seattle	0.08013	0.10278	HaLRTC
Portland	0.01878	0.05069	HaLRTC
City	0.08024	0.08209	TT-WOPT

ingness; evaluation uses clean missing entries. Baselines are PALM, CP-WOPT, HaLRTC, LRTC-TNN, and, where applicable, TT-WOPT or LATC. Validation uses 10% of observed entries to select configurations before refitting. CBO selects $(R, \lambda) = (8, 10^{-3})$ on CAVE and $(4, 10^{-4})$ elsewhere. Tuning, loading, and metric computation are excluded from solver time. Complete-sweep checkpoints target five seconds, rather than imposing an exact common stopping time.

Table 1 reports clean 80%-missing results. CBO achieves the lowest mean RMSE in nine of ten clean dataset–missingness settings; the exception is Bus at 80% missingness. Across the reported noise and missingness conditions, it wins 36 of 40 cells. Figure 6 uses three shared masks and plots actual median times at nominal 1–10 second targets. Rankings change with the available computation, so the five-second comparison describes a finite-budget operating point. These results are retained from the supplied benchmark; its optimized solver implementations and full trial records are not included in this archive, so the real-data comparison has not been independently rerun. Appendix I gives additional completion results and diagnostics.

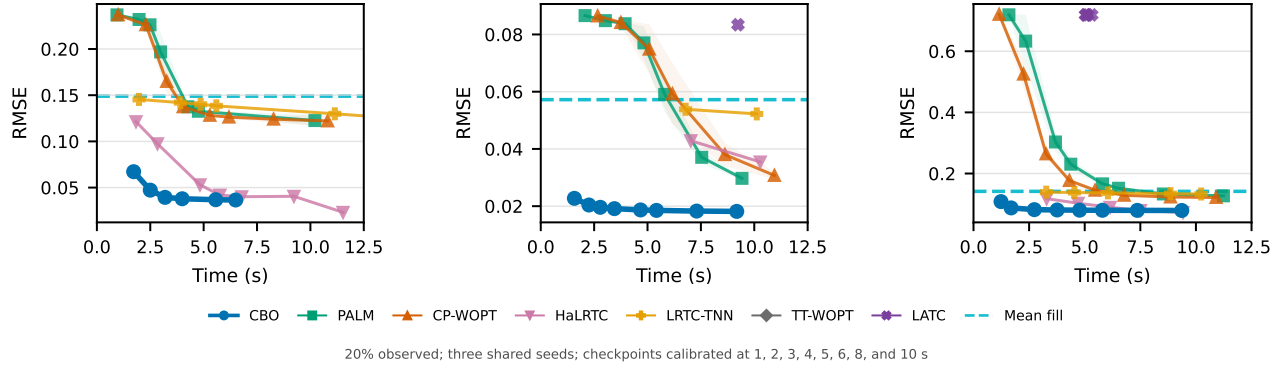


Figure 6: Time-accuracy trade-off on CAVE Balloons, Portland, and Seattle (left to right), with clean observations and 80% missingness. Mean missing-entry RMSE over three shared masks is plotted against actual median solver time at complete-sweep checkpoints targeting 1–10 seconds. Timing excludes tuning, loading, and metric evaluation. Dashed lines denote mean filling. Figure 10 in the supplement shows all five datasets.

Table 2: Matched-objective optimizer ablation at 32 sweeps: mean missing-entry RMSE / median solver seconds over five masks.

Update	CAVE	City
Exact cyclic	0.0366 /0.706	0.0654 /6.025
Gradient cyclic	0.1306/0.151	0.1274/1.490
Hybrid cyclic	0.1138/0.537	0.0816/3.739
PALM	0.1306/0.251	0.1274/2.641

6.5 Benefit of Exact Block Updates

The optimizer ablation holds the objective, observations, initial factors, $R = 8$, and $\lambda = 10^{-3}$ fixed. It uses a $64 \times 64 \times 31$ CAVE tensor and a $64 \times 80 \times 3 \times 48$ City tensor, both with 90% missingness. Table 2 compares exact, gradient, and hybrid cyclic updates with PALM after 32 sweeps; all statistics use five shared seeds. Exact updates cost more per sweep but reach substantially lower reconstruction errors. On CAVE, their normalized gradient norm is 0.0424, versus 0.964 for gradient cyclic/PALM; on City, the corresponding values are 1.14 and 44.3. These finite checkpoints show progress, not convergence of the runs. The full trajectories and an accelerated comparator, which is outside Theorem 1, are in the supplement.

6.6 Scaling with Tensor Size and Width

The supplied scaling study uses fresh processes to measure peak memory. Over 0.13–8.5 million entries, exact cyclic runtime has fitted log-log exponents 1.031–1.049 for orders three through five at fixed width and observation rate. Width changes are more costly: at 2.10 million entries and 10% observations, increasing R from four to 32 raises exact-sweep time from 0.211 to 8.689 seconds and peak memory from 165 to 1162 MB.

These measurements are consistent with (19), without establishing a universal scaling law. The supplement gives observation-rate sweeps, full curves, and gradient-method comparisons.

7 DISCUSSION & CONCLUSION

Width controls both capacity and regularization: redundant components can lower the penalty for an unchanged tensor. Data scale also matters, since ϕ_N mixes two amplitude powers for $N > 2$. Joint selection of R and λ , under consistent normalization, is therefore essential. The diagnostics answer different questions. Gap certificates measure agreement with tensor norms, held-out error measures recovery, and full gradient measures stationarity. Balancing excess $\mathcal{R} - \sum_r \phi_N(\mu_r)$ isolates removable scaling imbalance, but its vanishing alone does not certify stationarity. Smaller relative gaps at different fitted tensors need not imply better recovery. The convergence theorem guarantees stationarity at fixed width, not global optimality. Included scripts reproduce the synthetic norm and width studies; supplied real-data comparisons have not been independently rerun. The evidence separates efficient optimization, norm approximation, and recovery. Statistical guarantees and fitted real-data gap measurements remain open.

AI USE STATEMENT

Generative AI tools assisted with mathematical derivation checks, proof exposition, computational-complexity analysis, experiment-code development and review, numerical checks, figure preparation, and manuscript editing. The authors are responsible for verifying the final content and conclusions.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes]
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes]
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Not Applicable]
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Yes]
 - (b) Complete proofs of all theoretical results. [Yes]
 - (c) Clear explanations of any assumptions. [Yes]
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Partial: synthetic gap and rank experiments are included; real-data benchmark code and complete settings remain to be supplied.]
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes]
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes]
- (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes]
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (a) Citations of the creator If your work uses existing assets. [Yes]
 - (b) The license information of the assets, if applicable. [Not Applicable]
 - (c) New assets either in the supplemental material or as a URL, if applicable. [Yes: synthetic gap and rank experiment code and generated results are included.]
 - (d) Information about consent from data providers/curators. [Not Applicable]
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable]
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. [Not Applicable]
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable]
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable]

Scalable Low-Rank Tensor Completion: Supplementary Materials

A FULL FORMULATION AND NORM-GAP ANALYSIS

This supplement provides complete statements, proofs, and protocols for the results summarized in the main paper. The detailed notation and numbered identities also support the accompanying verification scripts.

A.1 Matrix completion and Burer–Monteiro factorization

The matrix nuclear norm gives the convex objective

$$\min_{\mathbf{X}} \frac{1}{2} \|P_{\Omega}(\mathbf{X} - \mathbf{Y})\|_F^2 + \lambda \|\mathbf{X}\|_*. \quad (20)$$

For a matrix $\mathbf{X}$ and any factor width $r \geq \text{rank}(\mathbf{X})$, the Burer–Monteiro variational identity is

$$\|\mathbf{X}\|_* = \min_{\mathbf{A}\mathbf{B}^\top = \mathbf{X}} \frac{1}{2} (\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2). \quad (21)$$

If $\mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\top$ is an SVD, equality is attained by $\mathbf{A} = \mathbf{U}\Sigma^{1/2}$ and $\mathbf{B} = \mathbf{V}\Sigma^{1/2}$. The factorized matrix objective is therefore an exact variational reformulation when the factor width is sufficient:

$$F(\mathbf{A}, \mathbf{B}) = \frac{1}{2} \|P_{\Omega}(\mathbf{Y} - \mathbf{A}\mathbf{B}^\top)\|_F^2 + \frac{\lambda}{2} (\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2). \quad (22)$$

Theorem 2. *For every matrix $\mathbf{X}$,*

$$\|\mathbf{X}\|_* = \min_{\mathbf{A}\mathbf{B}^\top = \mathbf{X}} \frac{1}{2} (\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2).$$

If $\mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\top$ is an SVD, an optimal pair is $\mathbf{A} = \mathbf{U}\Sigma^{1/2}$ and $\mathbf{B} = \mathbf{V}\Sigma^{1/2}$.

Thus, with a sufficient factor width, minimizing the matrix factor objective in Eq. (22) is equivalent to the nuclear-norm problem in Eq. (20).

A.2 Tensor norms and CP-factor surrogate

For tensors of order greater than two, there is no single canonical nuclear norm. We use the atomic tensor nuclear norm as our principal reference. For $\mathcal{X} \in \mathbb{R}^{I_1 \times \dots \times I_N}$, define the set of unit rank-one atoms

$$\mathfrak{A}_{\text{at}} := \{\mathbf{u}_1 \circ \dots \circ \mathbf{u}_N : \mathbf{u}_i \in \mathbb{R}^{I_i}, \|\mathbf{u}_i\|_2 = 1, i = 1, \dots, N\}. \quad (23)$$

The atomic tensor nuclear norm is

$$\|\mathcal{X}\|_{\text{at}} := \inf_{R \in \{1, 2, \dots\}, \lambda_r \in \mathbb{R}, \mathbf{u}_r \in \mathfrak{A}_{\text{at}} (r=1, \dots, R), \mathcal{X} = \sum_{r=1}^R \lambda_r \mathbf{u}_r} \sum_{r=1}^R |\lambda_r|. \quad (24)$$

Thus the infimum ranges over all finite decompositions of $\mathcal{X}$ into unit rank-one atoms. The corresponding atomic-norm completion problem is

$$\min_{\mathcal{X}} \frac{1}{2} \|P_{\Omega}(\mathcal{X} - \mathcal{Y})\|_F^2 + \lambda \|\mathcal{X}\|_{\text{at}}. \quad (25)$$

A distinct alternative is the average unfolding nuclear norm

$$\|\mathcal{X}\|_{\text{unf}} := \frac{1}{N} \sum_{i=1}^N \|\mathbf{X}_{(i)}\|_*. \quad (26)$$

It gives the separate convex completion problem

$$\min_{\mathcal{X}} \frac{1}{2} \|P_{\Omega}(\mathcal{X} - \mathcal{Y})\|_F^2 + \lambda \|\mathcal{X}\|_{\text{unf}}. \quad (27)$$

These two tensor penalties are not interchangeable. We use the atomic norm for the gap analysis below and treat the unfolding norm as a separate quantity that the same factor penalty also upper-bounds.

A rank- $\tilde{R}$ CP factorization is

$$\mathcal{X} = \sum_{r=1}^{\tilde{R}} \mathbf{a}_{1,r} \circ \cdots \circ \mathbf{a}_{N,r} = \llbracket \mathbf{A}_1, \dots, \mathbf{A}_N \rrbracket, \quad (28)$$

where $\mathbf{A}_i = [\mathbf{a}_{i,1}, \dots, \mathbf{a}_{i,\tilde{R}}] \in \mathbb{R}^{I_i \times \tilde{R}}$. For mode i , let

$$\mathbf{B}_i := \mathbf{A}_N \odot \cdots \odot \mathbf{A}_{i+1} \odot \mathbf{A}_{i-1} \odot \cdots \odot \mathbf{A}_1.$$

Then $\mathbf{X}_{(i)} = \mathbf{A}_i \mathbf{B}_i^\top$, and the Khatri–Rao Gram identity is

$$\|\mathbf{B}_i\|_F^2 = \text{tr} \left(\underset{j \neq i}{*} \mathbf{A}_j^\top \mathbf{A}_j \right), \quad (29)$$

where $*$ is the Hadamard product. Define the CP-factor regularizer

$$\mathcal{R}(\mathbf{A}_1, \dots, \mathbf{A}_N) := \frac{1}{2N} \sum_{i=1}^N \left[\|\mathbf{A}_i\|_F^2 + \text{tr} \left(\underset{j \neq i}{*} \mathbf{A}_j^\top \mathbf{A}_j \right) \right]. \quad (30)$$

Proposition 3 (Upper bounds for the two tensor norms). *For every CP factorization $\mathcal{X} = \llbracket \mathbf{A}_1, \dots, \mathbf{A}_N \rrbracket$,*

$$\|\mathcal{X}\|_{\text{at}} \leq \mathcal{R}(\mathbf{A}_1, \dots, \mathbf{A}_N), \quad \|\mathcal{X}\|_{\text{unf}} \leq \mathcal{R}(\mathbf{A}_1, \dots, \mathbf{A}_N). \quad (31)$$

Proof. Write $u_{i,r} = \|\mathbf{a}_{i,r}\|_2$ and $\mu_r = \prod_{i=1}^N u_{i,r}$. Normalizing each nonzero component in (28) gives a feasible atomic decomposition, so $\|\mathcal{X}\|_{\text{at}} \leq \sum_r \mu_r$. For every i, r , the scalar inequality

$$\frac{1}{2} \left(u_{i,r}^2 + \prod_{j \neq i} u_{j,r}^2 \right) \geq \mu_r$$

then yields the first bound. For the second, the matrix factorization inequality $\|\mathbf{A}_i \mathbf{B}_i^\top\|_* \leq \frac{1}{2} (\|\mathbf{A}_i\|_F^2 + \|\mathbf{B}_i\|_F^2)$ holds for every factor pair. Applying it to the CP-constrained pair $(\mathbf{A}_i, \mathbf{B}_i)$, summing over modes, and using (29) gives the second bound. $\square$

The matrix factorization inequality gives an upper bound after the right factor is restricted to a Khatri–Rao product; it does not give an equality for that restricted problem. No later result relies on such an equality. For $N = 2$, $\mathcal{R}$ reduces to $\frac{1}{2} (\|\mathbf{A}_1\|_F^2 + \|\mathbf{A}_2\|_F^2)$ and the matrix variational identity is recovered when the factor width is sufficient. For $N > 2$, minimizing over a fixed-width CP factorization is generally a nonconvex surrogate for either tensor-norm problem above, not an equivalent reformulation.

We minimize the following objective over $\mathbf{A}_1, \dots, \mathbf{A}_N$:

$$F(\mathbf{A}_1, \dots, \mathbf{A}_N) := \frac{1}{2} \|P_{\Omega}(\mathcal{Y} - \llbracket \mathbf{A}_1, \dots, \mathbf{A}_N \rrbracket)\|_F^2 + \lambda \mathcal{R}(\mathbf{A}_1, \dots, \mathbf{A}_N). \quad (32)$$

A.3 Tightness and Gap Analysis

We next characterize the gap between the atomic tensor nuclear norm in (24) and the surrogate regularizer. This analysis separates the total approximation error into two components: the error associated with the chosen CP decomposition and the error induced by the scaling of the factor vectors.

Let $\mathcal{X} = \sum_{k=1}^{\tilde{R}} v_{1,k} \circ v_{2,k} \circ \dots \circ v_{N,k}$, where $A_i = [v_{i,1}, \dots, v_{i,\tilde{R}}]$, $u_{i,k} := \|v_{i,k}\|_2$, $\mu_k := \prod_{i=1}^N u_{i,k}$. For every nonzero component, define the normalized vectors $\hat{v}_{i,k} := \frac{v_{i,k}}{u_{i,k}}$. For a zero-weight component ($\mu_k = 0$), choose arbitrary unit vectors $\hat{v}_{i,k}$; that component contributes zero to the sum. Then

$$\mathcal{X} = \sum_{k=1}^{\tilde{R}} \mu_k \hat{v}_{1,k} \circ \dots \circ \hat{v}_{N,k}, \quad \|\hat{v}_{i,k}\|_2 = 1.$$

The surrogate regularizer is the function defined in Eq. (30).

The following result gives an exact decomposition of the gap.

Proposition 4 (Exact decomposition of the surrogate gap). *For any CP factorization $\mathcal{X} = [[A_1, \dots, A_N]]$, the surrogate regularizer satisfies*

$$\mathcal{R}(A_1, \dots, A_N) = \sum_{k=1}^{\tilde{R}} \mu_k + \frac{1}{2N} \sum_{k=1}^{\tilde{R}} \sum_{i=1}^N \left(u_{i,k} - \prod_{j \neq i} u_{j,k} \right)^2. \quad (33)$$

Consequently,

$$\mathcal{R}(A_1, \dots, A_N) - \|\mathcal{X}\|_{\text{at}} = \Delta_{\text{dec}} + \Delta_{\text{scale}}, \quad (34)$$

where

$$\Delta_{\text{dec}} := \sum_{k=1}^{\tilde{R}} \mu_k - \|\mathcal{X}\|_{\text{at}} \geq 0, \quad \Delta_{\text{scale}} := \frac{1}{2N} \sum_{k=1}^{\tilde{R}} \sum_{i=1}^N \left(u_{i,k} - \prod_{j \neq i} u_{j,k} \right)^2 \geq 0. \quad (35)$$

We leave proof to appendix to save space. Proposition 4 shows that the total gap has two distinct sources. The decomposition gap Δ_{dec} measures the suboptimality of the chosen normalized CP representation relative to a nuclear-norm-minimizing representation. The scaling gap Δ_{scale} measures the additional looseness introduced by the quadratic inequality used to construct the surrogate.

A.3.1 Optimal scaling of each CP component

The CP representation is invariant to rescaling of its factor vectors. We next characterize the smallest possible surrogate value among all equivalent scalings of a fixed rank-one component.

Consider a component with coefficient $\mu > 0$, and let its factor norms satisfy $\prod_{i=1}^N u_i = \mu$. Its contribution to the surrogate is

$$q_N(u_1, \dots, u_N) = \frac{1}{2N} \sum_{i=1}^N \left(u_i^2 + \frac{\mu^2}{u_i^2} \right). \quad (36)$$

Proposition 5 (Minimum gap under balanced scaling). *For a fixed coefficient $\mu > 0$,*

$$\min_{\substack{u_i > 0 \\ \prod_{i=1}^N u_i = \mu}} q_N(u_1, \dots, u_N) = \phi_N(\mu) := \frac{1}{2} \left(\mu^{2/N} + \mu^{2-2/N} \right). \quad (37)$$

The minimum is attained at the balanced scaling

$$u_1 = \dots = u_N = \mu^{1/N}. \quad (38)$$

The corresponding minimum scaling gap is

$$g_N(\mu) := \phi_N(\mu) - \mu = \frac{1}{2} \left(\mu^{1/N} - \mu^{1-1/N} \right)^2. \quad (39)$$

We leave the proof to appendix.

The relative scaling gap can equivalently be written as

$$\frac{g_N(\mu)}{\mu} = \cosh \left(\left(1 - \frac{2}{N} \right) \log \mu \right) - 1. \quad (40)$$

For the matrix case $N = 2$, we have $g_2(\mu) = 0$ for every $\mu > 0$, which recovers the exact matrix Burer–Monteiro variational identity. For tensors of order $N > 2$, $g_N(\mu) = 0 \iff \mu = 1$. Therefore, even after optimally balancing the factor norms, the surrogate is generally not exact for higher-order tensors.

For coefficients close to one, a Taylor expansion gives

$$\frac{g_N(\mu)}{\mu} = \frac{1}{2} \left(1 - \frac{2}{N} \right)^2 (\log \mu)^2 + \mathcal{O}((\log \mu)^4). \quad (41)$$

This expansion controls only the scaling gap. The total gap is small only when the decomposition gap is also small; concentration of the coefficient magnitudes alone does not establish tightness.

Corollary 1 (Automatic balancing at stationary points). *Let $N \geq 2$ and $\lambda > 0$. At every unconstrained stationary point of F in (32), each component r satisfies*

$$\|\mathbf{a}_{1,r}\|_2 = \cdots = \|\mathbf{a}_{N,r}\|_2 = \mu_r^{1/N},$$

where $\mu_r := \prod_i \|\mathbf{a}_{i,r}\|_2$. Thus every component is either entirely zero or nonzero in every mode. Extending ϕ_N and g_N continuously by $\phi_N(0) = g_N(0) = 0$, we have

$$\mathcal{R} = \sum_r \phi_N(\mu_r), \quad \Delta_{\text{scale}} = \sum_r g_N(\mu_r). \quad (42)$$

The proof is in Appendix E. Hence the stationary points reached by the convergence theorem attain the best scaling for their component magnitudes. The remaining scaling gap is intrinsic to those magnitudes; this result does not imply that the decomposition gap vanishes.

A.3.2 Expected gap under random CP coefficients

We next provide a conditional expectation analysis. Assume the normalized CP representation attains the infimum in the atomic-norm definition almost surely, so that $\|\mathcal{X}\|_{\text{at}} = \sum_{k=1}^{\tilde{R}} \mu_k$. Assume further that the CP coefficients $M_1, \dots, M_{\tilde{R}}$ are independent and identically distributed positive random variables, and that every rank-one component is represented using the balanced scaling in (38). Then

$$\mathbb{E}[\mathcal{R}(A_1, \dots, A_N) - \|\mathcal{X}\|_{\text{at}}] = \tilde{R} \mathbb{E}[g_N(M)]. \quad (43)$$

Using (39), we obtain

$$\mathbb{E}[g_N(M)] = \frac{1}{2} \mathbb{E}[M^{2/N}] + \frac{1}{2} \mathbb{E}[M^{2-2/N}] - \mathbb{E}[M], \quad (44)$$

provided that the corresponding moments are finite.

A convenient model for positive CP coefficients is the lognormal distribution. Suppose $\log M \sim \mathcal{N}(\eta, \sigma_\mu^2)$. Since $\mathbb{E}[M^p] = \exp\left(p\eta + \frac{p^2 \sigma_\mu^2}{2}\right)$, the expected gap per component, with $c := 1 - 2/N$ and $v := \sigma_\mu^2$, is

$$\mathbb{E}[g_N(M)] = e^{\eta+v/2} \left[e^{c^2 v/2} \cosh(c(\eta+v)) - 1 \right]. \quad (45)$$

In particular, when $\eta = 0$ and σ_μ^2 is small,

$$\mathbb{E}[g_N(M)] = \frac{1}{2} \left(1 - \frac{2}{N} \right)^2 \sigma_\mu^2 + \mathcal{O}((\sigma_\mu^2)^2). \quad (46)$$

For fixed order $N > 2$, the leading small-variance term grows with the variance of the log-coefficients. Its coefficient is zero at $N = 2$ and increases toward $1/2$ as N grows. This conditional calculation concerns the scaling gap and does not bound the decomposition gap of an arbitrary fitted CP representation.

Computable interval for a fitted factorization. For a fitted CP representation, set $U_{\text{CP}} := \sum_{r=1}^{\tilde{R}} \mu_r$. The atomic norm is bounded below by every unfolding nuclear norm, while $\|\mathcal{X}\|_{\text{at}} \leq U_{\text{CP}} \leq \mathcal{R}$. Hence

$$\mathcal{R} - U_{\text{CP}} \leq \mathcal{R} - \|\mathcal{X}\|_{\text{at}} \leq \mathcal{R} - \max_{1 \leq i \leq N} \|\mathbf{X}_{(i)}\|_*. \quad (47)$$

The lower endpoint is exactly the scaling gap; the upper endpoint requires evaluating the mode-unfolding nuclear norms. We evaluate this interval for generated CP factors in Section I.6 and for fitted synthetic completion factors in Section 6.3. Corresponding intervals for fitted real-data factors remain to be evaluated.

A.4 Entrywise Gaussian tensors: spectral and atomic-norm bounds

This subsection gives a finite-sample benchmark for an entrywise Gaussian tensor. The entries themselves are assumed independent. Gaussian CP coefficients or Gaussian factor entries do not, in general, imply this assumption: the resulting tensor entries can be dependent.

Let $N \geq 2$, define $D := \prod_{i=1}^N I_i$, and write

$$I_{\max} := \max_i I_i, \quad S := \sum_{i=1}^N \sqrt{I_i}.$$

For some $\sigma > 0$, suppose that all entries satisfy

$$G_{i_1, \dots, i_N} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad 1 \leq i_j \leq I_j. \quad (48)$$

Define the tensor spectral (injective) norm by

$$\|\mathcal{G}\|_{\text{sp}} := \sup_{\|\mathbf{u}_i\|_2=1, i=1, \dots, N} |\langle \mathcal{G}, \mathbf{u}_1 \circ \dots \circ \mathbf{u}_N \rangle|. \quad (49)$$

Because the atom set in Eq. (23) is symmetric, $\|\cdot\|_{\text{sp}}$ is the dual norm of $\|\cdot\|_{\text{at}}$. We use the following known bound (Hu, 2015, Lemma 5.1), with a short fiber-decomposition proof included for completeness.

Lemma 1 (Fiber upper bound). *Every $\mathcal{H} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ satisfies*

$$\|\mathcal{H}\|_{\text{at}} \leq \sqrt{\frac{D}{I_{\max}}} \|\mathcal{H}\|_F. \quad (50)$$

Proof. Choose a mode m with $I_m = I_{\max}$ and let $J = D/I_{\max}$. Write the tensor as the sum of its J mode- m fibers embedded in the tensor space. Each summand has the fiber vector $\mathbf{v}_\alpha$ in mode m and standard basis vectors in the other modes. A nonzero summand is therefore a unit rank-one atom multiplied by $\|\mathbf{v}_\alpha\|_2$. The atomic decomposition and Cauchy-Schwarz imply

$$\|\mathcal{H}\|_{\text{at}} \leq \sum_{\alpha=1}^J \|\mathbf{v}_\alpha\|_2 \leq \sqrt{J} \left(\sum_{\alpha=1}^J \|\mathbf{v}_\alpha\|_2^2 \right)^{1/2} = \sqrt{J} \|\mathcal{H}\|_F.$$

□

Proposition 6 (Gaussian spectral and atomic-norm bounds). *Under model (48), for every $t \geq 0$,*

$$\mathbb{P}\{\|\mathcal{G}\|_{\text{sp}} > \sigma(S + t)\} \leq e^{-t^2/2}. \quad (51)$$

Moreover, for $0 < \delta < 1/2$, set

$$x_\delta := \log(2/\delta), \quad (52)$$

$$L_\delta := \frac{\sigma[D - 2\sqrt{Dx_\delta}]_+}{S + \sqrt{2x_\delta}}, \quad (53)$$

$$U_\delta := \sigma \sqrt{\frac{D}{I_{\max}}} \left(\sqrt{D} + \sqrt{2 \log(1/\delta)} \right), \quad (54)$$

where $[a]_+ := \max\{a, 0\}$. Then

$$\mathbb{P}\{L_\delta \leq \|\mathcal{G}\|_{\text{at}} \leq U_\delta\} \geq 1 - 2\delta. \quad (55)$$

In particular,

$$(1 - \delta)L_\delta \leq \mathbb{E}\|\mathcal{G}\|_{\text{at}} \leq \frac{\sigma D}{\sqrt{I_{\max}}}. \quad (56)$$

Proof. We first bound the expected spectral norm. Index the product of unit spheres by $\mathbf{u} = (\mathbf{u}_1, \dots, \mathbf{u}_N)$ and set

$$X_{\mathbf{u}} := \langle \mathcal{G}, \mathbf{u}_1 \circ \dots \circ \mathbf{u}_N \rangle.$$

Let $\mathbf{g}_i \sim \mathcal{N}(0, I_{I_i})$ be independent standard Gaussian vectors and define $Y_{\mathbf{u}} := \sum_{i=1}^N \langle \mathbf{g}_i, \mathbf{u}_i \rangle$. These are centered Gaussian processes. For two indices $\mathbf{u}, \mathbf{w}$, write $a_i := \langle \mathbf{u}_i, \mathbf{w}_i \rangle \in [-1, 1]$. Then

$$\mathbb{E}(X_{\mathbf{u}} - X_{\mathbf{w}})^2 = 2\sigma^2 \left(1 - \prod_{i=1}^N a_i\right) \leq 2\sigma^2 \sum_{i=1}^N (1 - a_i) = \sigma^2 \mathbb{E}(Y_{\mathbf{u}} - Y_{\mathbf{w}})^2.$$

The inequality follows from the telescoping identity

$$1 - \prod_{i=1}^N a_i = \sum_{i=1}^N \left(\prod_{j < i} a_j \right) (1 - a_i).$$

Each prefix product is at most one and $1 - a_i \geq 0$, so the comparison also holds when some inner products are negative. Since changing the sign of one factor preserves the index set, $\sup_{\mathbf{u}} X_{\mathbf{u}} = \|\mathcal{G}\|_{\text{sp}}$. The Sudakov–Fernique comparison inequality applies to these continuous processes on a compact index set and gives

$$\mathbb{E}\|\mathcal{G}\|_{\text{sp}} = \mathbb{E} \sup_{\mathbf{u}} X_{\mathbf{u}} \leq \sigma \mathbb{E} \sup_{\mathbf{u}} Y_{\mathbf{u}} = \sigma \sum_{i=1}^N \mathbb{E}\|\mathbf{g}_i\|_2 \leq \sigma S,$$

where the last step is Jensen’s inequality. This comparison is the tensor analogue of the usual Gaussian matrix operator-norm bound; see, for example, (Yang, 2021).

The spectral norm is 1-Lipschitz with respect to the Frobenius norm: the triangle inequality gives

$$|\|\mathcal{H}\|_{\text{sp}} - \|\mathcal{K}\|_{\text{sp}}| \leq \|\mathcal{H} - \mathcal{K}\|_F.$$

It is therefore σ -Lipschitz as a function of the standardized Gaussian entries. Gaussian concentration around its expectation proves (51).

By duality and the choice $\mathcal{Y} = \mathcal{G}/\|\mathcal{G}\|_{\text{sp}}$,

$$\|\mathcal{G}\|_{\text{at}} \geq \frac{\|\mathcal{G}\|_F^2}{\|\mathcal{G}\|_{\text{sp}}} \quad (57)$$

whenever $\mathcal{G} \neq 0$, an event of probability one under model (48). Since $\|\mathcal{G}\|_F^2/\sigma^2 \sim \chi_D^2$, the Laurent–Massart lower-tail inequality (Laurent and Massart, 2000) gives

$$\mathbb{P}\{\|\mathcal{G}\|_F^2 < \sigma^2[D - 2\sqrt{Dx_\delta}]_+\} \leq \delta/2.$$

When the positive part vanishes, this statement is trivial. Equation (51) also gives

$$\mathbb{P}\{\|\mathcal{G}\|_{\text{sp}} > \sigma(S + \sqrt{2x_\delta})\} \leq \delta/2.$$

A union bound and (57) prove $\mathbb{P}\{\|\mathcal{G}\|_{\text{at}} \geq L_\delta\} \geq 1 - \delta$.

For the upper bound, $\|\mathcal{G}\|_F$ is σ -Lipschitz as a function of the standardized entries and satisfies $\mathbb{E}\|\mathcal{G}\|_F \leq \sigma\sqrt{D}$. Thus Gaussian concentration yields

$$\mathbb{P}\left\{\|\mathcal{G}\|_F > \sigma\left(\sqrt{D} + \sqrt{2\log(1/\delta)}\right)\right\} \leq \delta.$$

Lemma 1 now implies $\mathbb{P}\{\|\mathcal{G}\|_{\text{at}} > U_\delta\} \leq \delta$. Combining the two tail events proves Eq. (55). Taking expectations in (50) gives the expectation upper bound; the lower bound follows from the lower-tail event of probability at least $1 - \delta$ and nonnegativity. $\square$

For fixed N and $I_i = n$, the spectral norm has order $\sigma\sqrt{n}$ in expectation: the upper bound is $N\sigma\sqrt{n}$, while a matching lower order follows by fixing unit vectors in all but one mode and maximizing over the remaining mode. The resulting Gaussian vector has expected Euclidean norm of order $\sigma\sqrt{n}$. For the atomic norm, at any fixed $0 < \delta < 1/2$, as $n \rightarrow \infty$,

$$L_\delta \sim \frac{\sigma}{N} n^{N-1/2}, \quad U_\delta \sim \sigma n^{N-1/2}.$$

Thus the bounds establish the dimension order $\sigma n^{N-1/2}$ for the atomic norm at fixed confidence and for its expectation, with constants depending on N (and, for the probability statement, the confidence level). They do not identify a sharp leading constant. For $N = 2$, the spectral and atomic norms are exactly the matrix operator and nuclear norms, respectively.

For an exact CP representation $\mathcal{G} = \llbracket \mathbf{A}_1, \dots, \mathbf{A}_N \rrbracket$, write

$$\rho_R := \mathcal{R}(\mathbf{A}_1, \dots, \mathbf{A}_N), \quad \Delta_{\text{sur}} := \rho_R - \|\mathcal{G}\|_{\text{at}}.$$

On the event in (55), Proposition 3 implies

$$[\rho_R - U_\delta]_+ \leq \Delta_{\text{sur}} \leq \rho_R - L_\delta. \quad (58)$$

This transfers a bound on the atomic norm to the gap for every exact CP representation of the realized tensor, including a data-dependent one. It does not by itself bound ρ_R or the expected gap; those depend on how the CP factors are selected. In a completion problem, the result applies to the target only when that target has the entrywise-i.i.d. Gaussian model above; it does not automatically apply to a fitted completion.

For independent entries $G_{i_1, \dots, i_N} \sim \mathcal{N}(\mu, \sigma^2)$ with known mean μ , the atomic-norm probability interval remains valid after replacing L_δ by $[L_\delta - |\mu|\sqrt{D}]_+$ and U_δ by $U_\delta + |\mu|\sqrt{D}$. The gap interval then uses the same adjusted endpoints. This follows from the triangle inequality, since the all-ones rank-one tensor has atomic norm $\sqrt{D}$. Marginal normality alone, without independence or a specified dependence structure, is insufficient for these bounds.

A.5 Gap analysis relative to the average unfolding norm

For $N > 2$, the atomic norm and the average unfolding norm are generally distinct tensor regularizers. Here we analyze the gap between the CP-factor regularizer $\mathcal{R}$ in (30) and the average unfolding norm $\|\cdot\|_{\text{unf}}$ in (26). The gap has an exact modewise expression in terms of Gram matrices.

For a CP factorization with $\tilde{R}$ columns, $\mathcal{X} = \llbracket \mathbf{A}_1, \dots, \mathbf{A}_N \rrbracket$, write $\rho_{\mathcal{R}} := \mathcal{R}(\mathbf{A}_1, \dots, \mathbf{A}_N)$ and define

$$\mathbf{B}_i := \mathbf{A}_N \odot \dots \odot \mathbf{A}_{i+1} \odot \mathbf{A}_{i-1} \odot \dots \odot \mathbf{A}_1, \quad \mathbf{P}_i := \mathbf{A}_i^\top \mathbf{A}_i, \quad \mathbf{Q}_i := \mathbf{B}_i^\top \mathbf{B}_i = \bigstar_{j \neq i} (\mathbf{A}_j^\top \mathbf{A}_j).$$

Thus $\mathbf{X}_{(i)} = \mathbf{A}_i \mathbf{B}_i^\top$. For positive semidefinite matrices $\mathbf{P}, \mathbf{Q}$ of the same size, define the squared Bures–Wasserstein distance (Bhatia et al., 2019)

$$d_{\text{BW}}^2(\mathbf{P}, \mathbf{Q}) := \text{tr}(\mathbf{P}) + \text{tr}(\mathbf{Q}) - 2 \text{tr}[(\mathbf{P}^{1/2} \mathbf{Q} \mathbf{P}^{1/2})^{1/2}]. \quad (59)$$

Proposition 7 (Exact unfolding-norm gap). *For every CP factorization $\mathcal{X} = \llbracket \mathbf{A}_1, \dots, \mathbf{A}_N \rrbracket$,*

$$\rho_{\mathcal{R}} - \|\mathcal{X}\|_{\text{unf}} = \frac{1}{2N} \sum_{i=1}^N d_{\text{BW}}^2(\mathbf{P}_i, \mathbf{Q}_i) \geq 0. \quad (60)$$

Equality holds if and only if $\mathbf{P}_i = \mathbf{Q}_i$ for every mode i .

Proof. Let $\mathbf{C}_i := \mathbf{B}_i \mathbf{P}_i^{1/2}$. Then

$$(\mathbf{A}_i \mathbf{B}_i^\top)^\top (\mathbf{A}_i \mathbf{B}_i^\top) = \mathbf{C}_i \mathbf{C}_i^\top, \quad \mathbf{C}_i^\top \mathbf{C}_i = \mathbf{P}_i^{1/2} \mathbf{Q}_i \mathbf{P}_i^{1/2}.$$

The nonzero eigenvalues of $\mathbf{C}_i \mathbf{C}_i^\top$ and $\mathbf{C}_i^\top \mathbf{C}_i$ coincide. Taking their square roots and summing gives

$$\|\mathbf{A}_i \mathbf{B}_i^\top\|_* = \text{tr}[(\mathbf{P}_i^{1/2} \mathbf{Q}_i \mathbf{P}_i^{1/2})^{1/2}]. \quad (61)$$

The definition of $\mathcal{R}$ and the Khatri–Rao Gram identity give

$$\rho_{\mathcal{R}} = \frac{1}{2N} \sum_{i=1}^N [\text{tr}(\mathbf{P}_i) + \text{tr}(\mathbf{Q}_i)], \quad \|\mathbf{X}\|_{\text{unf}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{A}_i \mathbf{B}_i^\top\|_*.$$

Substitution proves (60). The orthogonal Procrustes identity also gives

$$d_{\text{BW}}^2(\mathbf{P}, \mathbf{Q}) = \min_{\mathbf{U} \in \mathbb{O}(\tilde{R})} \|\mathbf{P}^{1/2} - \mathbf{Q}^{1/2} \mathbf{U}\|_F^2,$$

where $\mathbb{O}(\tilde{R})$ is the orthogonal group. This identity holds also for singular positive semidefinite matrices. Its minimum is zero if and only if $\mathbf{P} = \mathbf{Q}$: equality of the square-root factors up to a right orthogonal matrix implies equality of their products with their transposes, and the converse follows by taking $\mathbf{U} = \mathbf{I}$. This establishes nonnegativity and the equality condition. $\square$

The gap in (60) measures, mode by mode, the mismatch between the Gram matrix of that mode’s CP factors and the Gram matrix of the Khatri–Rao product of the remaining factors. It is a matrix-factorization balance gap; it has no additional CP-decomposition gap of the form in the atomic-norm analysis. For a single CP component ($\tilde{R} = 1$), writing $u_i := \|\mathbf{a}_{i,1}\|_2$ gives

$$d_{\text{BW}}^2(\mathbf{P}_i, \mathbf{Q}_i) = \left(u_i - \prod_{j \neq i} u_j \right)^2.$$

Thus the formula reduces to the componentwise scaling gap.

A.5.1 Gaussian-entry bounds

Suppose the tensor entries are independent centered Gaussians, as in (48). Let $D := \prod_{i=1}^N I_i$, $J_i := D/I_i$, and $r_i := \min\{I_i, J_i\}$. For $0 < \delta < 1/2$, define

$$x_\delta := \log(2/\delta), \quad a_\delta := [D - 2\sqrt{Dx_\delta}]_+, \quad s_\delta := \sqrt{2\log(2N/\delta)}, \quad q_\delta := \sqrt{2\log(1/\delta)}. \quad (62)$$

Here $[a]_+ := \max\{a, 0\}$. Set

$$L_{\text{unf},\delta} := \frac{\sigma a_\delta}{N} \sum_{i=1}^N \frac{1}{\sqrt{I_i} + \sqrt{J_i} + s_\delta}, \quad (63)$$

$$U_{\text{unf},\delta} := \frac{\sigma}{N} \left(\sum_{i=1}^N \sqrt{r_i} \right) (\sqrt{D} + q_\delta). \quad (64)$$

Proposition 8 (Gaussian unfolding-norm bounds). *Under model (48),*

$$\mathbb{P}\{L_{\text{unf},\delta} \leq \|\mathbf{G}\|_{\text{unf}} \leq U_{\text{unf},\delta}\} \geq 1 - 2\delta. \quad (65)$$

The expectation satisfies

$$(1 - \delta)L_{\text{unf},\delta} \leq \mathbb{E}\|\mathbf{G}\|_{\text{unf}} \leq \frac{\sigma\sqrt{D}}{N} \sum_{i=1}^N \sqrt{r_i}. \quad (66)$$

Proof. Each unfolding $\mathbf{G}_{(i)}$ is an $I_i \times J_i$ matrix with independent $\mathcal{N}(0, \sigma^2)$ entries, and $\|\mathbf{G}_{(i)}\|_F = \|\mathbf{G}\|_F$. The matrix case of (51) implies that

$$\|\mathbf{G}_{(i)}\|_{\text{op}} \leq \sigma(\sqrt{I_i} + \sqrt{J_i} + s_\delta)$$

holds with probability at least $1 - \delta/(2N)$. A union bound makes these bounds simultaneous with probability at least $1 - \delta/2$; no independence among unfoldings is required. The Laurent–Massart inequality gives

$$\mathbb{P}\{\|\mathbf{G}\|_F^2 < \sigma^2 a_\delta\} \leq \delta/2.$$

When $a_\delta = 0$, this statement is trivial. For any nonzero matrix $\mathbf{M}$, its singular values satisfy

$$\|\mathbf{M}\|_* \geq \frac{\|\mathbf{M}\|_F^2}{\|\mathbf{M}\|_{\text{op}}}.$$

Applying this inequality to each unfolding on the preceding events and averaging over modes yields $\|\mathcal{G}\|_{\text{unf}} \geq L_{\text{unf},\delta}$ with probability at least $1 - \delta$.

For the upper bound, Cauchy–Schwarz on the singular values gives $\|\mathbf{G}_{(i)}\|_* \leq \sqrt{r_i} \|\mathcal{G}\|_F$. As a function of standard Gaussian entries, $\|\mathcal{G}\|_F$ is σ -Lipschitz, and $\mathbb{E}\|\mathcal{G}\|_F \leq \sigma\sqrt{D}$. Gaussian concentration therefore yields

$$\mathbb{P}\left\{\|\mathcal{G}\|_F > \sigma(\sqrt{D} + q_\delta)\right\} \leq \delta.$$

Averaging the unfolding inequalities proves the upper bound with failure probability at most δ . A union bound gives (65). Taking expectations in the same inequalities gives the expectation upper bound. The lower bound follows from the lower-tail event of probability at least $1 - \delta$ and nonnegativity of the norm. $\square$

If $I_i = n$ for every mode and N is fixed, then $J_i = n^{N-1}$ and $r_i = n$. As $n \rightarrow \infty$ at fixed confidence level, both endpoints in (65) have order $\sigma n^{(N+1)/2}$. Hence both the high-probability and expectation bounds have sharp dimension dependence in this balanced, fixed-order setting, although their finite-sample constants can be conservative. Together with (56), this yields

$$\frac{\mathbb{E}\|\mathcal{G}\|_{\text{at}}}{\mathbb{E}\|\mathcal{G}\|_{\text{unf}}} = \Theta_N\left(n^{(N-2)/2}\right).$$

Here the constants in Θ_N may depend on the fixed order N . This ratio of expectations quantifies the separation between the two reference norms for entrywise Gaussian tensors. For $N = 2$, the norms coincide and this ratio is exactly one.

For any exact CP factorization of the realized tensor $\mathcal{G}$, use $\rho_{\mathcal{R}} := \mathcal{R}(\mathbf{A}_1, \dots, \mathbf{A}_N)$ and set

$$\Delta_{\text{unf}} := \rho_{\mathcal{R}} - \|\mathcal{G}\|_{\text{unf}}.$$

Proposition 3 and the event in (65) give

$$[\rho_{\mathcal{R}} - U_{\text{unf},\delta}]_+ \leq \Delta_{\text{unf}} \leq \rho_{\mathcal{R}} - L_{\text{unf},\delta}. \quad (67)$$

This holds with probability at least $1 - 2\delta$, simultaneously for all exact CP representations of the realized tensor, including a data-dependent choice. It does not bound $\rho_{\mathcal{R}}$ or the expected gap without a model for how the factors are selected. It applies to a completion target only if that target has the stated entrywise-independent Gaussian distribution; it does not automatically apply to a fitted completion.

If instead the entries are independent $\mathcal{N}(\mu, \sigma^2)$ variables with a common known mean μ , write $\mathcal{G} = \mathcal{Z} + \mu\mathbf{1}$ with centered $\mathcal{Z}$. The same probability bound holds with endpoints

$$L_{\text{unf},\delta}^{(\mu)} := [L_{\text{unf},\delta} - |\mu|\sqrt{D}]_+, \quad U_{\text{unf},\delta}^{(\mu)} := U_{\text{unf},\delta} + |\mu|\sqrt{D}.$$

Indeed, each unfolding of the all-ones tensor has nuclear norm $\sqrt{D}$, so $\|\mathbf{1}\|_{\text{unf}} = \sqrt{D}$; the claim follows from the triangle and reverse triangle inequalities. Substituting these shifted endpoints in (67) gives the corresponding gap interval for an exact CP representation of $\mathcal{G}$.

B OPTIMIZATION AND CONVERGENCE ANALYSIS

Throughout this section, $N \geq 2$, the factor width $1 \leq R < \infty$ is fixed, and $\lambda > 0$. The objective in Eq. (32) is generally nonconvex jointly in the factors. Fixing all factors except $\mathbf{A}_i$ gives a strongly convex quadratic block problem. We optimize these blocks cyclically and establish convergence to stationary points.

Let $\mathcal{A} = (\mathbf{A}_1, \dots, \mathbf{A}_N)$ and $\|\mathcal{A}\|^2 = \sum_i \|\mathbf{A}_i\|_F^2$. Define

$$\mathbf{B}_i := \mathbf{A}_N \odot \dots \odot \mathbf{A}_{i+1} \odot \mathbf{A}_{i-1} \odot \dots \odot \mathbf{A}_1. \quad (68)$$

For $i \neq j$, let

$$h(i, j) := \underset{\ell \notin \{i, j\}}{*} \mathbf{A}_\ell^\top \mathbf{A}_\ell, \quad (69)$$

where an empty Hadamard product is the $R \times R$ all-ones matrix. Dropping terms independent of $\mathbf{A}_i$ gives the reduced block objective

$$\tilde{F}_i(\mathbf{A}_i) = \frac{1}{2} \|P_{\Omega_i}(\mathbf{Y}_{(i)} - \mathbf{A}_i \mathbf{B}_i^\top)\|_F^2 + \frac{\lambda}{2N} \|\mathbf{A}_i\|_F^2 + \frac{\lambda}{2N} \sum_{j \neq i} \text{tr}(h(i, j) * (\mathbf{A}_i^\top \mathbf{A}_i)). \quad (70)$$

Here P_{Ω_i} is the mode- i observation mask. With $\mathbf{a}_i = \text{vec}(\mathbf{A}_i)$, define

$$\begin{aligned} \mathbf{y}_i &:= \text{vec}(\mathbf{Y}_{(i)}), & \mathbf{P}_i &:= P_{\Omega_{v,i}}, & \mathbf{K}_i &:= \mathbf{B}_i \otimes \mathbf{I}_{I_i}, \\ \mathbf{D}_i &:= \text{Diag} \left(\text{diag} \left(\sum_{j \neq i} h(i, j) \right) \right), & \mathbf{M}_i &:= \mathbf{D}_i \otimes \mathbf{I}_{I_i}. \end{aligned} \quad (71)$$

The operator diag extracts a matrix diagonal, whereas Diag forms a diagonal matrix from a vector. The vectorized reduced objective is

$$\tilde{F}_i(\mathbf{a}_i) = \frac{1}{2} \|\mathbf{P}_i(\mathbf{y}_i - \mathbf{K}_i \mathbf{a}_i)\|_2^2 + \frac{\lambda}{2N} \mathbf{a}_i^\top (\mathbf{I}_{I_i R} + \mathbf{M}_i) \mathbf{a}_i. \quad (72)$$

Retaining the terms independent of $\mathbf{a}_i$ gives the actual restriction of the global objective:

$$F_i(\mathbf{a}_i; \mathcal{A}_{-i}) = \tilde{F}_i(\mathbf{a}_i) + c_i(\mathcal{A}_{-i}) = F(\mathbf{A}_1, \dots, \mathbf{A}_N). \quad (73)$$

Define

$$\mathbf{H}_i := \mathbf{K}_i^\top \mathbf{P}_i^\top \mathbf{P}_i \mathbf{K}_i + \frac{\lambda}{N} (\mathbf{I}_{I_i R} + \mathbf{M}_i), \quad \mathbf{b}_i := \mathbf{K}_i^\top \mathbf{P}_i^\top \mathbf{P}_i \mathbf{y}_i. \quad (74)$$

Then

$$\nabla_i F(\mathcal{A}) = \mathbf{H}_i \mathbf{a}_i - \mathbf{b}_i, \quad \nabla_{ii}^2 F(\mathcal{A}) = \mathbf{H}_i. \quad (75)$$

For a symmetric idempotent mask, $\mathbf{P}_i^\top \mathbf{P}_i = \mathbf{P}_i$.

Lemma 2 (Block smoothness and strong convexity). *For fixed $\mathcal{A}_{-i}$, the objective (73) is strongly convex with modulus at least $\mu := \lambda/N$. Its exact block Lipschitz constant is $L_i^{\text{exact}} = \lambda_{\max}(\mathbf{H}_i)$. A computable bound is*

$$L_i^{\text{exact}} \leq \hat{L}_i := \lambda_{\max} \left(\underset{\ell \neq i}{*} \mathbf{A}_\ell^\top \mathbf{A}_\ell \right) + \frac{\lambda}{N} (1 + \|\mathbf{D}_i\|_2). \quad (76)$$

Proof. Since $\mathbf{P}_i^\top \mathbf{P}_i \succeq 0$ and $\mathbf{M}_i \succeq 0$, $\mathbf{H}_i \succeq (\lambda/N) \mathbf{I}_{I_i R}$. Also,

$$\lambda_{\max}(\mathbf{K}_i^\top \mathbf{P}_i^\top \mathbf{P}_i \mathbf{K}_i) = \|\mathbf{P}_i \mathbf{K}_i\|_2^2 \leq \|\mathbf{K}_i\|_2^2 = \|\mathbf{B}_i\|_2^2. \quad (77)$$

The Khatri–Rao identity gives $\mathbf{B}_i^\top \mathbf{B}_i = \underset{\ell \neq i}{*} \mathbf{A}_\ell^\top \mathbf{A}_\ell$, and $\|\mathbf{M}_i\|_2 = \|\mathbf{D}_i\|_2$, proving the bound. $\square$

A bound using only column norms. Set $q_{\ell r} := \|\mathbf{a}_{\ell, r}\|_2^2$. The diagonal entries of $\mathbf{D}_i$ are

$$d_{ir} = \sum_{j \neq i} \prod_{\ell \notin \{i, j\}} q_{\ell r}. \quad (78)$$

An optional, potentially looser bound is

$$\tilde{L}_i := \sum_{r=1}^R \prod_{\ell \neq i} q_{\ell r} + \frac{\lambda}{N} \left(1 + \max_r d_{ir} \right). \quad (79)$$

Indeed, the first term equals $\|\mathbf{B}_i\|_F^2 \geq \|\mathbf{B}_i\|_2^2$, so $\tilde{L}_i \geq \hat{L}_i \geq L_i^{\text{exact}}$. For each r , both this product and d_{ir} can be computed without division: initialize $(P, S) = (1, 0)$, visit each $\ell \neq i$, and update

$$(P, S) \leftarrow (P q_{\ell r}, S q_{\ell r} + P)$$

simultaneously using the old P, S . After the last update, $P = \prod_{\ell \neq i} q_{\ell r}$ and $S = d_{ir}$. Thus, computing $\mathbf{D}_i$ and $\tilde{L}_i$ costs $\mathcal{O}(NR)$ per block, including when some column norms vanish. Initialize the column norms once and refresh those of each updated factor in $\mathcal{O}(I_i R)$.

Observed-entry block updates. The Kronecker matrix $\mathbf{K}_i$ and Hessian $\mathbf{H}_i$ are algebraic notation; an implementation need not construct them. For an observed coordinate $\omega = (\omega_1, \dots, \omega_N)$, define

$$\mathbf{b}_\omega^{(i)} := \odot_{j \neq i} \mathbf{A}_j[\omega_j, :]^\top \in \mathbb{R}^R.$$

The prediction is $\hat{y}_\omega = \langle \mathbf{A}_i[\omega_i, :]^\top, \mathbf{b}_\omega^{(i)} \rangle$. After grouping variables by row, the data Hessian is block diagonal. For $s \in [I_i]$, let

$$\mathbf{G}_{i,s} := \sum_{\omega \in \Omega: \omega_i = s} \mathbf{b}_\omega^{(i)} (\mathbf{b}_\omega^{(i)})^\top, \quad \mathbf{c}_{i,s} := \sum_{\omega \in \Omega: \omega_i = s} y_\omega \mathbf{b}_\omega^{(i)}. \quad (80)$$

The exact update solves the positive-definite row systems

$$\left[\mathbf{G}_{i,s} + \frac{\lambda}{N} (\mathbf{I}_R + \mathbf{D}_i) \right] \mathbf{A}_i[s, :]^\top = \mathbf{c}_{i,s}.$$

Each system has dimension R and can be solved by Cholesky factorization without forming an inverse.

A gradient update can instead accumulate the row gradients directly:

$$\mathbf{g}_{i,s} = \sum_{\omega \in \Omega: \omega_i = s} (\hat{y}_\omega - y_\omega) \mathbf{b}_\omega^{(i)} + \frac{\lambda}{N} (\mathbf{I}_R + \mathbf{D}_i) \mathbf{A}_i[s, :]^\top. \quad (81)$$

This avoids assembling $\mathbf{G}_{i,s}$. The current block is held fixed while its gradient is accumulated. Given a valid $L_i \geq \lambda_{\max}(\mathbf{H}_i)$, update

$$\mathbf{a}_i^+ = \mathbf{a}_i - \frac{1}{L_i} \nabla_i F(\mathcal{A}) = \mathbf{a}_i - \frac{1}{L_i} (\mathbf{H}_i \mathbf{a}_i - \mathbf{b}_i). \quad (82)$$

Either (76) or (79) is a valid choice.

Complexity of a streamed implementation. Let $M = |\Omega|$. Consider an implementation that processes observations in fixed-size batches and recomputes $\mathbf{b}_\omega^{(i)}$ from the currently available factors within each block. Forming all complements then costs $\mathcal{O}(N^2 MR)$ per sweep, respecting the cyclic order. Using the norm recurrence above, diagonal computations cost $\mathcal{O}(N^2 R)$ and column-norm refreshes cost $\mathcal{O}(R \sum_i I_i)$ per sweep. Accumulating row Hessians and solving their systems gives an exact-sweep bound

$$\mathcal{O}\left(N^2 MR + NMR^2 + N^2 R + R^3 \sum_i I_i\right).$$

Here initialization of the row systems and norm refreshes are absorbed by $R^3 \sum_i I_i$, since $R \geq 1$. A matrix-free gradient sweep using (79) costs

$$\mathcal{O}\left(N^2 MR + N^2 R + R \sum_i I_i\right).$$

If the full-Gram bound (76) is evaluated instead, forming and refreshing the factor Gram matrices, forming their Hadamard products, and computing exact eigenvalues add, respectively,

$$\mathcal{O}\left(R^2 \sum_i I_i + N^2 R^2 + NR^3\right)$$

per sweep, with $\mathcal{O}(NR^2)$ additional storage for cached Grams. These are alternative implementations of the same admissible update rule; the norm-only choice is not required by the convergence theorem.

With the norm-only construction, storing the observations and factors and accumulating systems for one mode at a time requires

$$\mathcal{O}\left(NM + R \sum_i I_i + \max_i I_i R^2\right)$$

memory for exact updates. Matrix-free gradient updates require $\mathcal{O}(NM + R \sum_i I_i)$ memory. These conservative bounds use no cache of all mode-complement products. They exclude dense ground-truth tensors, hyperparameter search, and metric computation.

Algorithm 2 Cyclic Block Optimization for Tensor Completion

Input: Observations $\{(\omega, y_\omega) : \omega \in \Omega\}$, factor width R , $\lambda > 0$, block types $q_i \in \{\text{AM}, \text{GD}\}$

1. Initialize $\mathcal{A}^0$ and its column norms; set $k \leftarrow 0$.
 2. **While** the stopping criterion is not satisfied:
 - (a) **For** $i = 1, \dots, N$, in order:
 - Compute $\mathbf{D}_i$ by (78).
 - **If** $q_i = \text{AM}$, accumulate $\mathbf{G}_{i,s}, \mathbf{c}_{i,s}$ by (80) and solve the row systems by Cholesky factorization.
 - **Otherwise**, accumulate the gradient by (81), choose a valid L_i , and apply (82).
 - Refresh the column norms of $\mathbf{A}_i$.
 - (b) Set $\mathcal{A}^{k+1}$ to the updated factors; increment $k \leftarrow k + 1$.
 3. **Return** $\hat{\mathcal{X}} = \llbracket \mathbf{A}_1^k, \dots, \mathbf{A}_N^k \rrbracket$.
-

B.1 Cyclic update rule

Let $\mathcal{A}^k$ denote the factors at the beginning of sweep k . The intermediate iterates are

$$\mathcal{A}^{k,i} := (\mathbf{A}_1^{k+1}, \dots, \mathbf{A}_i^{k+1}, \mathbf{A}_{i+1}^k, \dots, \mathbf{A}_N^k), \quad \mathcal{A}^{k,0} := \mathcal{A}^k, \quad \mathcal{A}^{k,N} := \mathcal{A}^{k+1}. \quad (83)$$

Each block therefore uses the latest values of the preceding blocks.

Choosing AM for every block gives cyclic exact minimization, and choosing GD for every block gives cyclic block-gradient descent. Hybrid choices are also allowed. We call the all-AM configuration CBO in the experiments.

B.2 Convergence analysis

The following result gives descent, stationary-point convergence, and a sublinear stationarity bound. It does not require joint convexity.

For each sweep k and block i , define

$$H_i^k := \nabla_{\text{vec}(\mathbf{A}_i)}^2 F(\mathcal{A}) \Big|_{\mathcal{A}=\mathcal{A}^{k,i-1}}.$$

Thus, H_i^k is the Hessian of the block- i restriction of F , evaluated immediately before updating $\mathbf{A}_i$, with $\mathbf{A}_j = \mathbf{A}_j^{k+1}$ for $j < i$ and $\mathbf{A}_j = \mathbf{A}_j^k$ for $j > i$. Since this restriction is quadratic in $\mathbf{A}_i$, its Hessian is independent of the value of $\mathbf{A}_i$.

Theorem 3 (Convergence of cyclic exact and gradient updates). *Fix $N \geq 2$, finite $R \geq 1$, and $\lambda > 0$. For gradient updates, assume $\lambda_{\max}(\mathbf{H}_i^k) \leq L_i^k \leq \bar{L} < \infty$, where $\bar{L} > 0$; both (76) and (79) satisfy this condition. For all-exact updates, $\bar{L}$ may be any positive constant.*

Let $\{\mathcal{A}^k\}$ be the completed-sweep iterates of Algorithm 2, with either update type in each block. Set $F_{\inf} := \inf_{\mathcal{A}} F(\mathcal{A})$, $\mu := \lambda/N$, and $E_0 := F(\mathcal{A}^0) - F_{\inf}$. Then:

1. The objective decreases monotonically, and

$$F(\mathcal{A}^k) - F(\mathcal{A}^{k+1}) \geq \frac{\mu}{2} \|\mathcal{A}^{k+1} - \mathcal{A}^k\|^2. \quad (84)$$

2. The iterates are bounded, and

$$\sum_{k=0}^{\infty} \|\mathcal{A}^{k+1} - \mathcal{A}^k\|^2 \leq \frac{2E_0}{\mu}, \quad \|\mathcal{A}^{k+1} - \mathcal{A}^k\| \longrightarrow 0. \quad (85)$$

3. Every accumulation point is stationary for F .

4. For every integer $K \geq 1$,

$$\min_{0 \leq k < K} \|\mathcal{A}^{k+1} - \mathcal{A}^k\|^2 \leq \frac{2E_0}{\mu K}. \quad (86)$$

5. Let Λ be a Lipschitz constant of ∇F on a compact convex set containing all iterates, including intermediate block iterates; such a constant exists because F is polynomial. Then

$$\min_{0 \leq k < K} \|\nabla F(\mathcal{A}^{k+1})\|^2 \leq \frac{4(\bar{L}^2 + N\Lambda^2)E_0}{\mu K}. \quad (87)$$

Consequently, the minimum gradient norm is $\mathcal{O}(K^{-1/2})$.

6. The complete sequence has finite length and converges to one stationary point.

Proof. Write $\Delta_i^k := \mathbf{a}_i^{k+1} - \mathbf{a}_i^k$. For an exact update, strong convexity and optimality of $\mathbf{a}_i^{k+1}$ give

$$F(\mathcal{A}^{k,i-1}) - F(\mathcal{A}^{k,i}) \geq \frac{\mu}{2} \|\mathbf{a}_i^{k+1} - \mathbf{a}_i^k\|_2^2 = \frac{\mu}{2} \|\Delta_i^k\|_2^2. \quad (88)$$

For a gradient update, block smoothness gives

$$F(\mathcal{A}^{k,i}) \leq F(\mathcal{A}^{k,i-1}) + \langle \nabla_i F(\mathcal{A}^{k,i-1}), \Delta_i^k \rangle + \frac{L_i^k}{2} \|\Delta_i^k\|_2^2.$$

Substituting $\Delta_i^k = -(L_i^k)^{-1} \nabla_i F(\mathcal{A}^{k,i-1})$ yields

$$F(\mathcal{A}^{k,i-1}) - F(\mathcal{A}^{k,i}) \geq \frac{\|\nabla_i F(\mathcal{A}^{k,i-1})\|_2^2}{2L_i^k} = \frac{L_i^k}{2} \|\Delta_i^k\|_2^2. \quad (89)$$

Since $L_i^k \geq \lambda_{\max}(\mathbf{H}_i^k) \geq \lambda_{\min}(\mathbf{H}_i^k) \geq \mu$, either update satisfies

$$F(\mathcal{A}^{k,i-1}) - F(\mathcal{A}^{k,i}) \geq \frac{\mu}{2} \|\Delta_i^k\|_2^2. \quad (90)$$

Each block changes exactly once in a sweep, so summing gives

$$F(\mathcal{A}^k) - F(\mathcal{A}^{k+1}) \geq \frac{\mu}{2} \sum_{i=1}^N \|\Delta_i^k\|_2^2 = \frac{\mu}{2} \|\mathcal{A}^{k+1} - \mathcal{A}^k\|^2. \quad (91)$$

This proves (84).

All regularization terms are nonnegative. In particular,

$$F(\mathcal{A}) \geq \frac{\lambda}{2N} \sum_{i=1}^N \|\mathbf{A}_i\|_F^2 \geq 0. \quad (92)$$

Consequently the initial level set $\mathcal{L}_0 := \{\mathcal{A} : F(\mathcal{A}) \leq F(\mathcal{A}^0)\}$ is compact. Blockwise descent keeps every intermediate and completed iterate in $\mathcal{L}_0$. Both computable bounds (76) and (79) are continuous functions of the factors, so their values on this compact set are uniformly bounded. This verifies the stated sufficient choices for the stepsize assumption without using that assumption to establish boundedness.

Summing (84) proves (85); averaging the first K terms proves (86). To relate steps to stationarity, define

$$\mathcal{Z}_i^k := \begin{cases} \mathcal{A}^{k,i}, & \text{for an exact update,} \\ \mathcal{A}^{k,i-1}, & \text{for a gradient update.} \end{cases}$$

Exact optimality gives $\nabla_i F(\mathcal{Z}_i^k) = 0$, whereas the gradient update gives $\nabla_i F(\mathcal{Z}_i^k) = -L_i^k \Delta_i^k$. Thus, in either case,

$$\|\nabla_i F(\mathcal{Z}_i^k)\|_2 \leq \bar{L} \|\Delta_i^k\|_2. \quad (93)$$

Because F is polynomial in the finitely many factor entries, its Hessian is continuous. The set $C := \text{conv}(\mathcal{L}_0)$ is compact and convex in finite dimensions. Hence

$$\Lambda := \max_{\mathcal{A} \in C} \|\nabla^2 F(\mathcal{A})\|_2 < \infty$$

is a Lipschitz constant for ∇F on C . This set contains every intermediate iterate and every line segment used below. Also, $\|\mathcal{A}^{k+1} - \mathcal{Z}_i^k\| \leq \|\mathcal{A}^{k+1} - \mathcal{A}^k\|$, since the former difference contains only a subset of the block changes in the latter. Therefore

$$\|\nabla_i F(\mathcal{A}^{k+1})\|_2 \leq \|\nabla_i F(\mathcal{Z}_i^k)\|_2 + \Lambda \|\mathcal{A}^{k+1} - \mathcal{Z}_i^k\| \leq \bar{L} \|\Delta_i^k\|_2 + \Lambda \|\mathcal{A}^{k+1} - \mathcal{A}^k\|. \quad (94)$$

Squaring with $(a+b)^2 \leq 2a^2 + 2b^2$ and summing over blocks gives

$$\|\nabla F(\mathcal{A}^{k+1})\|^2 \leq 2(\bar{L}^2 + N\Lambda^2) \cdot \|\mathcal{A}^{k+1} - \mathcal{A}^k\|^2. \quad (95)$$

Combining this inequality with (86) proves (87). Together with square summability, it gives $\nabla F(\mathcal{A}^{k+1}) \rightarrow 0$. Boundedness and continuity of ∇F then imply that every accumulation point is stationary.

For convergence of the entire sequence, we verify the conditions of Attouch et al. (2013, Theorem 2.9). The polynomial F is proper, continuous, and has the Kurdyka-Łojasiewicz property. Condition H1 (sufficient decrease) is (84) with $a = \mu/2 > 0$. Condition H2 (relative error) follows by taking square roots in (95), with $w^{k+1} = \nabla F(\mathcal{A}^{k+1})$ and $b = \sqrt{2(\bar{L}^2 + N\Lambda^2)} > 0$. Finally, compactness gives a subsequence $\mathcal{A}^{k_j} \rightarrow \bar{\mathcal{A}}$, and continuity gives $F(\mathcal{A}^{k_j}) \rightarrow F(\bar{\mathcal{A}})$, which is condition H3. The cited theorem therefore yields

$$\sum_{k=0}^{\infty} \|\mathcal{A}^{k+1} - \mathcal{A}^k\| < \infty$$

and convergence of $\mathcal{A}^k$ to a single stationary point. $\square$

For the all-gradient variant, define the intermediate stationarity measure

$$\mathcal{G}_k^2 := \sum_{i=1}^N \|\nabla_i F(\mathcal{A}^{k,i-1})\|_2^2. \quad (96)$$

Corollary 2 (Rate for cyclic block-gradient descent). *Under the assumptions of Theorem 3, if every block uses (82), then, for $K \geq 1$,*

$$\min_{0 \leq k < K} \mathcal{G}_k^2 \leq \frac{2\bar{L}(F(\mathcal{A}^0) - F_{\inf})}{K}. \quad (97)$$

Proof. By (89),

$$F(\mathcal{A}^k) - F(\mathcal{A}^{k+1}) \geq \sum_{i=1}^N \frac{\|\nabla_i F(\mathcal{A}^{k,i-1})\|_2^2}{2L_i^k} \geq \frac{\mathcal{G}_k^2}{2\bar{L}}. \quad (98)$$

Sum over $k = 0, \dots, K-1$ and take the minimum. $\square$

C PROOF OF PROPOSITION 4

Proof. Using $\|A_i\|_F^2 = \sum_{k=1}^{\tilde{R}} u_{i,k}^2$ and the identity $\text{tr}(*_{j \neq i} A_j^\top A_j) = \sum_{k=1}^{\tilde{R}} \prod_{j \neq i} u_{j,k}^2$, we obtain

$$\mathcal{R}(A_1, \dots, A_N) = \frac{1}{2N} \sum_{k=1}^{\tilde{R}} \sum_{i=1}^N \left(u_{i,k}^2 + \prod_{j \neq i} u_{j,k}^2 \right). \quad (99)$$

For each component k ,

$$\begin{aligned} \frac{1}{2N} \sum_{i=1}^N \left(u_{i,k}^2 + \prod_{j \neq i} u_{j,k}^2 \right) - \mu_k &= \frac{1}{2N} \sum_{i=1}^N \left[u_{i,k}^2 + \left(\prod_{j \neq i} u_{j,k} \right)^2 - 2u_{i,k} \prod_{j \neq i} u_{j,k} \right] \\ &= \frac{1}{2N} \sum_{i=1}^N \left(u_{i,k} - \prod_{j \neq i} u_{j,k} \right)^2. \end{aligned} \quad (100)$$

where we used $u_{i,k} \prod_{j \neq i} u_{j,k} = \prod_{j=1}^N u_{j,k} = \mu_k$. Summing (100) over k proves (33).

Moreover, the normalized CP representation $\mathcal{X} = \sum_{k=1}^{\tilde{R}} \mu_k \hat{v}_{1,k} \circ \cdots \circ \hat{v}_{N,k}$ is feasible in the definition of the tensor nuclear norm. Hence, $\|\mathcal{X}\|_{\text{at}} \leq \sum_{k=1}^{\tilde{R}} \mu_k$. Subtracting $\|\mathcal{X}\|_{\text{at}}$ from (33) gives (34). $\square$

D PROOF OF PROPOSITION 5

Proof. Let $x_i = u_i^2$. Since $\prod_{i=1}^N x_i = \mu^2$, the arithmetic-geometric mean inequality gives

$$\frac{1}{N} \sum_{i=1}^N x_i \geq \left(\prod_{i=1}^N x_i \right)^{1/N} = \mu^{2/N}.$$

Similarly,

$$\frac{1}{N} \sum_{i=1}^N \frac{\mu^2}{x_i} \geq \left(\prod_{i=1}^N \frac{\mu^2}{x_i} \right)^{1/N} = \mu^{2-2/N}.$$

Substituting these inequalities into (36) gives

$$q_N(u_1, \dots, u_N) \geq \frac{1}{2} \left(\mu^{2/N} + \mu^{2-2/N} \right).$$

Equality in both arithmetic-geometric mean inequalities holds if and only if

$$u_1^2 = \cdots = u_N^2 = \mu^{2/N},$$

which proves (38) and (37). Finally,

$$\phi_N(\mu) - \mu = \frac{1}{2} \left(\mu^{2/N} + \mu^{2-2/N} - 2\mu \right) = \frac{1}{2} \left(\mu^{1/N} - \mu^{1-1/N} \right)^2.$$

$\square$

E PROOF OF COROLLARY 1

Proof. Fix a component r and two distinct modes i, j , and let $s_\ell := \|\mathbf{a}_{\ell,r}\|_2^2$. Consider the curve $\mathbf{a}_{i,r}(t) = e^t \mathbf{a}_{i,r}$, $\mathbf{a}_{j,r}(t) = e^{-t} \mathbf{a}_{j,r}$, leaving all other factors unchanged. The represented tensor, and hence the loss, is constant along this curve, including when some factor vectors are zero. Differentiating the columnwise regularizer gives

$$\left. \frac{dF}{dt} \right|_{t=0} = \frac{\lambda}{N} (s_i - s_j) \left(1 + \prod_{\ell \notin \{i,j\}} s_\ell \right). \quad (101)$$

The empty product is one. At a stationary point the derivative vanishes. The parenthesized factor is strictly positive and $\lambda > 0$, so $s_i = s_j$. Since i, j are arbitrary, every mode has the same norm in component r . This also shows that if one factor is zero, all factors in that component are zero. Substitution into the columnwise formula for $\mathcal{R}$ and the definition of Δ_{scale} proves (42). $\square$

F MORE ON GAP ANALYSIS

F.1 Expected objective gap under random observations

Let

$$J_*(\mathcal{X}; \Omega) = \frac{1}{2} \|P_\Omega(\mathcal{Y} - \mathcal{X})\|_F^2 + \lambda \|\mathcal{X}\|_{\text{at}} \quad (102)$$

denote the original nuclear-norm-regularized objective, and let

$$J_{\mathcal{R}}(A_1, \dots, A_N; \Omega) = \frac{1}{2} \|P_\Omega(\mathcal{Y} - [[A_1, \dots, A_N]])\|_F^2 + \lambda \mathcal{R}(A_1, \dots, A_N). \quad (103)$$

For the same represented tensor $\mathcal{X} = [[A_1, \dots, A_N]]$, the data-fitting terms are identical. Therefore, for every observation set Ω ,

$$J_{\mathcal{R}}(A_1, \dots, A_N; \Omega) - J_*(\mathcal{X}; \Omega) = \lambda [\mathcal{R}(A_1, \dots, A_N) - \|\mathcal{X}\|_{\text{at}}]. \quad (104)$$

For fixed $\mathcal{Y}$ and factors $A_1, \dots, A_N$ chosen independently of the observation mask, if every tensor entry is observed independently with probability p , then

$$\mathbb{E}_{\Omega} [\|P_{\Omega}(\mathcal{Y} - \mathcal{X})\|_F^2] = p \|\mathcal{Y} - \mathcal{X}\|_F^2.$$

Consequently,

$$\mathbb{E}_{\Omega}[J_{\mathcal{R}}] = \frac{p}{2} \|\mathcal{Y} - \mathcal{X}\|_F^2 + \lambda \mathcal{R}(A_1, \dots, A_N), \quad (105)$$

$$\mathbb{E}_{\Omega}[J_*] = \frac{p}{2} \|\mathcal{Y} - \mathcal{X}\|_F^2 + \lambda \|\mathcal{X}\|_{\text{at}}, \quad (106)$$

and hence

$$\mathbb{E}_{\Omega}[J_{\mathcal{R}} - J_*] = \lambda [\mathcal{R}(A_1, \dots, A_N) - \|\mathcal{X}\|_{\text{at}}]. \quad (107)$$

Thus, random sampling affects the expected data-fitting term, but does not change the regularization gap when the factor matrices are fixed.

F.2 A computable interval for the unknown gap

Although the atomic tensor nuclear norm is generally difficult to evaluate, the gap can be bounded using quantities obtained directly from the learned factors. Define $U_{\text{CP}} := \sum_{k=1}^{\tilde{R}} \mu_k$. Then

$$\|\mathcal{X}\|_{\text{at}} \leq U_{\text{CP}} \leq \mathcal{R}(A_1, \dots, A_N).$$

Moreover, every matrix unfolding provides a lower bound:

$$\max_{1 \leq i \leq N} \|\mathcal{X}_{(i)}\|_* \leq \|\mathcal{X}\|_{\text{at}}.$$

Therefore,

$$\mathcal{R}(A_1, \dots, A_N) - U_{\text{CP}} \leq \mathcal{R}(A_1, \dots, A_N) - \|\mathcal{X}\|_{\text{at}} \leq \mathcal{R}(A_1, \dots, A_N) - \max_{1 \leq i \leq N} \|\mathcal{X}_{(i)}\|_*. \quad (108)$$

The lower endpoint is precisely the scaling gap Δ_{scale} , whereas the upper endpoint can be evaluated by computing the nuclear norms of the unfolding matrices. This interval is a computable diagnostic. Numerical gap intervals for the fitted real tensors are not reported in this paper.

G FACTOR WIDTH: ATTAINABLE PENALTIES AND ALGORITHMIC EFFECTS

Throughout this section, $N \geq 2$, R is a positive integer, and the factor matrices have R columns. The number of columns is an upper bound on CP rank, not a certificate of the tensor's minimal CP rank. Columns may vanish, be redundant, or cancel. We extend the balanced component penalty continuously to zero:

$$\phi_N(0) = 0, \quad \phi_N(\nu) = \frac{1}{2} \left(\nu^{2/N} + \nu^{2-2/N} \right) \quad (\nu > 0).$$

For any factorization, Proposition 5 gives

$$\mathcal{R}(\mathcal{A}) \geq \sum_{r=1}^R \phi_N(\mu_r), \quad \mu_r := \prod_{i=1}^N \|\mathbf{a}_{i,r}\|_2.$$

Zero components obey this bound as well, and can be replaced by all-zero columns when minimizing the penalty for a fixed tensor.

G.1 Existence and nesting

Proof of the existence and nesting assertions in Proposition 2. Since $\mathcal{R}(\mathcal{A}) \geq (2N)^{-1} \sum_i \|\mathbf{A}_i\|_F^2$, the continuous objective (32) is coercive when $\lambda > 0$. It therefore attains its global minimum at each finite width. The representation map $\mathcal{A} \mapsto [\mathbf{A}_1, \dots, \mathbf{A}_N]$ is polynomial, so its preimage of a fixed tensor is closed. If that preimage is nonempty, a minimizing sequence for $\mathcal{P}_R(\mathcal{X})$ has bounded factors by coercivity of $\mathcal{R}$. A convergent subsequence has a feasible limit, which attains $\mathcal{P}_R(\mathcal{X})$ by continuity.

Appending one all-zero column to every factor changes neither the represented tensor nor the regularizer. The latter assertion uses $N \geq 2$, so every new mode-complement column is also zero. Embedding a minimizing representation or a globally minimizing factor tuple into width $R+1$ proves both monotonicity inequalities. $\square$

In particular, nonclosedness of unrestricted CP-rank constraint sets does not invalidate this existence argument: a fixed finite penalty level bounds the factors themselves, not only the represented tensors.

A useful lower bound also holds at every feasible width. Let $t := \|\mathcal{X}\|_{\text{at}}$, $a := 2/N$, and $c := 1 - 2/N$. Then

$$\mathcal{P}_R(\mathcal{X}) \geq \max \left\{ t, \frac{1}{2} (t^a + R^{-c} t^{2-a}) \right\}. \quad (109)$$

Indeed, every factorization has total component amplitude $T := \sum_r \mu_r \geq t$. Since $0 < a \leq 1$ and $2 - a \geq 1$, subadditivity of x^a and Jensen's inequality give

$$\sum_r \mu_r^a \geq T^a, \quad \sum_r \mu_r^{2-a} \geq R^{a-1} T^{2-a}.$$

Combining these inequalities with balancing and $T \geq t$ gives the second lower bound; the first is the atomic-norm upper-bound property. For $N > 2$, the factor R^{-c} makes explicit how a finite width limits the reduction of the large-amplitude penalty. This bound is generally not asserted to be tight.

G.2 Exact reduction for a rank-one tensor

Proof of (14). Write $\mathcal{U} = \mathbf{u}_1 \circ \dots \circ \mathbf{u}_N$ with $\|\mathbf{u}_i\|_2 = 1$. Its atomic norm is one: the displayed representation gives the upper bound, while every unit atom has Frobenius norm one, implying $\|\mathcal{U}\|_{\text{at}} \geq \|\mathcal{U}\|_F = 1$. The case $\mu = 0$ follows from all-zero factors, so suppose $\mu > 0$.

For any width- R factorization of $\mu\mathcal{U}$, let $v_r = \prod_i \|\mathbf{a}_{i,r}\|_2$ and $T = \sum_r v_r$. Normalization of its nonzero components gives an atomic decomposition, so $T \geq \mu > 0$. Set $\nu_r = (\mu/T)v_r$. Then $\nu_r \geq 0$ and $\sum_r \nu_r = \mu$. Because ϕ_N is increasing on $[0, \infty)$,

$$\mathcal{P}_R(\mathcal{A}) \geq \sum_r \phi_N(v_r) \geq \sum_r \phi_N(\nu_r) \geq \min_{\substack{z_r \geq 0 \\ \sum_r z_r = \mu}} \sum_r \phi_N(z_r).$$

The scalar simplex is compact and ϕ_N is continuous, so its minimum is attained. Conversely, for any feasible scalar allocation $(\nu_1, \dots, \nu_R)$, choose $\mathbf{a}_{i,r} = \nu_r^{1/N} \mathbf{u}_i$ for every i, r . These balanced factors represent $\sum_r \nu_r \mathcal{U} = \mu\mathcal{U}$ and have penalty $\sum_r \phi_N(\nu_r)$. They attain the lower bound, proving the characterization. $\square$

For $N > 2$, $\phi_N(\nu) \geq \nu$, with equality precisely when $\nu = 0$ or $\nu = 1$. The allocation formula and attainment therefore imply

$$\mathcal{P}_R(\mu\mathcal{U}) = \mu \iff \mu \in \{0, 1, \dots, R\}.$$

When $\mu = m$ is a positive integer, m identical unit-amplitude components give equality as soon as $R \geq m$. When $\mu > 0$ is not an integer, the inequality is strict at every finite width. These statements concern rank-one tensors; no such sufficiency claim is made for general tensors merely from the value of their atomic norm. For $N = 2$, $\phi_2(\nu) = \nu$, and $\mathcal{P}_R(\mu\mathcal{U}) = \mu$ for every $R \geq 1$ and $\mu \geq 0$.

There is also no splitting benefit for a rank-one tensor of amplitude $0 < \mu \leq 1$, regardless of width:

$$\mathcal{P}_R(\mu\mathcal{U}) = \phi_N(\mu).$$

For $N > 2$, the ratio $\phi_N(\nu)/\nu = \cosh(c \log \nu)$ decreases as ν increases on $(0, 1]$. Every positive allocation weight is at most μ , hence $\sum_r \phi_N(\nu_r) \geq [\phi_N(\mu)/\mu] \sum_r \nu_r = \phi_N(\mu)$. A single nonzero component attains this value. The matrix case follows directly from $\phi_2(\nu) = \nu$.

G.3 Equal splitting as an explicit construction

For $\mu > 0$ and an integer $1 \leq k \leq R$, choose k allocation weights equal to μ/k and pad the remaining columns with zeros. The attained penalty is

$$k\phi_N(\mu/k) = \frac{1}{2} \left(\mu^{2/N} k^{1-2/N} + \mu^{2-2/N} k^{-(1-2/N)} \right) = \mu \cosh(c \log(\mu/k)), \quad c = 1 - \frac{2}{N}. \quad (110)$$

Its relative gap from the atomic norm is $\cosh(c \log(\mu/k)) - 1$. For $N > 2$, the best integer k within this equal-split family minimizes $|\log(\mu/k)|$ over $1 \leq k \leq R$. It is 1 for $\mu \leq 1$, R for $\mu \geq R$, and otherwise one of $\lfloor \mu \rfloor, \lceil \mu \rceil$. Between consecutive choices k and $k+1$, the transition occurs at $\mu = \sqrt{k(k+1)}$, with both choices tied there. This identifies only the optimum within the equal-split construction, not a closed form for the general scalar allocation problem. For $N = 2$ all these choices have the same penalty μ . The increase of this construction's penalty for $k > \mu$ does not contradict nesting of $\mathcal{P}_R$: increasing the permitted width does not require using every column.

G.4 Zero-column invariance and the meaning of rank experiments

Suppose a width- R stationary factor tuple is embedded into a larger width by all-zero new columns in every mode. The old-column gradients are unchanged: the regularizer is a sum over columns and the represented tensor is unchanged. The data gradient for each new mode- i column contains the product of the other $N-1$ zero mode vectors, hence vanishes. Its regularizer gradient also vanishes at zero. Thus the padded tuple remains stationary for every $N \geq 2$.

Such columns cannot activate under either proposed update. A gradient step leaves their zero gradients unchanged. For an exact block update, the corresponding column of $\mathbf{B}_i$ is zero, so it has zero data right-hand side and zero data-Hessian coupling to the old columns. The strictly positive diagonal regularization term forces its unique minimizer to be zero. Induction through blocks and sweeps proves invariance. Therefore, experiments intended to explore additional components must initialize them nonzero. Nested feasible sets alone do not force separately initialized local solvers to return monotone objectives as R increases.

Every finite-width objective satisfies the convergence theorem with block strong-convexity lower bound λ/N . This lower bound is independent of R , but the ambient dimension, initial level set, full-gradient Lipschitz constant, and the constants in the stationarity bound may depend on R . No rank-uniform iteration-complexity or statistical-recovery guarantee follows.

For the streamed implementation described in the optimization section, with $M = |\Omega|$, the exact and norm-only, matrix-free gradient sweep costs are, respectively,

$$\mathcal{O}\left(N^2MR + NMR^2 + N^2R + R^3 \sum_i I_i\right) \quad \text{and} \quad \mathcal{O}\left(N^2MR + N^2R + R \sum_i I_i\right).$$

Storing the factors requires $\mathcal{O}(R \sum_i I_i)$ memory, with additional active-mode row Hessians of size $\mathcal{O}(\max_i I_i R^2)$ for exact updates. These computational effects, the nested representable classes, and the width-dependent induced penalty are distinct influences on performance. The convergence results alone establish none of the comparative recovery outcomes.

For a synthetic ground truth $\mathcal{X}_* = \sum_{r=1}^{r_*} \gamma_r \mathbf{u}_{1,r} \circ \cdots \circ \mathbf{u}_{N,r}$, with nonzero coefficients and orthonormal columns within every mode, its actual CP rank is exactly r_* . Indeed, its representation provides the upper bound, and each mode unfolding has matrix rank r_* , providing the matching lower bound. Thus $R < r_*$ excludes the exact full tensor, although its observed entries might still be fitted. In a general fitted representation, counting numerically nonzero components is a measure of active columns, not a certified CP-rank computation.

H CONTROLLED STUDY OF FACTOR WIDTH

This study isolates the role of the prescribed CP factor width in synthetic completion. It uses a new, independently checked implementation of exact cyclic minimization of the paper's objective; it does not reproduce the historical real-data solver or compare its speed against competing methods. The experiment directory includes all generated data, masks, selected factors, individual trajectories, and scripts.

Ground truth and paired observations. For each of five seeds, we construct a $16 \times 16 \times 16$ tensor $\mathcal{X}_\star = \sum_{r=1}^4 w_r u_{1,r} \otimes u_{2,r} \otimes u_{3,r}$, where $w = (8, 6, 4, 2)$ and each mode’s four columns are obtained by a thin QR factorization of an independent Gaussian matrix. Each unfolding has matrix rank four, so the CP rank is exactly four. Independent Gaussian noise has standard deviation $0.05 \|\mathcal{X}_\star\|_F / \sqrt{4096}$. A random permutation allocates a fixed 10% validation set, nested 20% and 40% training sets, and a common remaining approximately 50% test set (counts are rounded down). Thus tensor, noise, initialization seeds, and evaluation entries are paired across widths and observation rates. Entries used only in the larger training set are left unused in the smaller setting. Training and validation use noisy observations. Test error is $\|P_{\text{test}}(\hat{\mathcal{X}} - \mathcal{X}_\star)\|_F / \|P_{\text{test}}\mathcal{X}_\star\|_F$ and is never used for tuning.

Widths, regularization, and initialization. We fit $R \in \{1, 2, 4, 6, 8, 12\}$ using $\lambda = \alpha \|P_{\text{train}}\mathcal{Y}\|_F^2 / 20$ for $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}\}$; the divisor 20 is a fixed scale constant. For every width and regularization value, two initially balanced, nonzero Gaussian factor sets are used. Column directions are nested across widths, while the represented tensor’s Frobenius norm is rescaled to 25% of the training-only norm estimate $\sqrt{\|P_{\text{train}}\mathcal{Y}\|_F^2 / p_{\text{train}}}$. These choices avoid the absorbing zero-column initialization and a larger initial tensor scale at larger widths. For each λ , we retain the start with lower final training objective; we then select the regularization value by the noisy validation MSE. There are 360 fits and 60 selected models.

Exact updates and common stopping rule. For mode i , the implementation forms the masked normal equations for every row of A_i and solves them with a batched positive-definite dense linear solve. A separate verification script checks objective values, gradients, Hessians, and sufficient decrease against an independently constructed Khatri–Rao/observed-coordinate implementation, including zero columns and overcomplete widths. We monitor the full factor gradient and objective after every sweep, and stop when

$$\frac{\|\nabla F(A^k)\|_F}{\max\{1, \|\nabla F(A^0)\|_F\}} \leq 10^{-5}$$

or after 1,000 sweeps. The gradient norm concatenates all mode blocks. The one-seed, 300-sweep pilot was used only to assess runtime and stopping; the final budget was increased without changing the data, tuning grid, or selection rule. All capped runs remain in the summaries. Of the 60 selected fits, 31 meet the tolerance and 29 reach the cap. Recovery measurements are therefore finite-budget results, not a claim that every selected fit is stationary.

Timing and uncertainty. We fix OpenBLAS, OMP, and MKL thread settings to one. Wall-clock fitting times include objective and gradient monitoring, but exclude data creation, post-fit diagnostics, and CSV writes. We report both the selected run’s time and the full cost of all six tuning candidates. The reference implementation uses dense masks and tensors; its small-problem timings should not be extrapolated to a large sparse implementation. Error bars in the main recovery/timing panels are means and sample standard deviations over five paired seeds, not confidence intervals. Additional diagnostics below show the individual replicates as faint dots.

Fitted norm gaps and active columns. For each selected fit let $\rho = \mathcal{R}(A)$, $a = \sum_r \prod_i \|a_{i,r}\|_2$, and $b = \max_i \|\hat{X}_{(i)}\|_\star$. We compute the certified interval $[1 - a/\rho, 1 - b/\rho]$ for the normalized atomic gap and the exact normalized average-unfolding gap $1 - \|\hat{\mathcal{X}}\|_{\text{unf}}/\rho$. The gap figure averages the two interval endpoints; its shaded region is not a statistical error band or an estimate of an exact atomic norm. The exact unfolding gap can exceed the atomic interval’s upper endpoint. We also report the balancing excess $(\rho - \sum_r \phi_3(\mu_r))/\rho$ and the count of columns whose μ_r exceeds $10^{-3} \max_s \mu_s$. This active-column count does not determine the CP rank, since dependent components and cancellations remain possible. Widths below four cannot represent the complete true tensor; widths above four enlarge the feasible family but need not improve validation-selected, finite-budget test error.

Table 3: Width study: test relative error and cost per sweep (mean $\pm$ standard deviation over five paired seeds). The last column counts selected fits that met the gradient tolerance; the remaining fits reached the common 1,000-sweep cap.

Observed	R	Test relative error	Sweep time (ms)	Converged
20%	1	0.791 ± 0.202	0.59 ± 0.06	4/5
20%	2	0.522 ± 0.176	1.05 ± 0.11	3/5
20%	4	0.033 ± 0.006	1.18 ± 0.05	1/5
20%	6	0.051 ± 0.004	1.31 ± 0.19	1/5
20%	8	0.064 ± 0.006	1.42 ± 0.22	2/5
20%	12	0.088 ± 0.016	2.25 ± 0.16	4/5
40%	1	0.716 ± 0.078	0.60 ± 0.08	2/5
40%	2	0.468 ± 0.101	1.05 ± 0.06	4/5
40%	4	0.018 ± 0.001	1.13 ± 0.06	0/5
40%	6	0.027 ± 0.002	1.24 ± 0.06	1/5
40%	8	0.031 ± 0.003	1.38 ± 0.11	4/5
40%	12	0.033 ± 0.002	2.12 ± 0.27	5/5

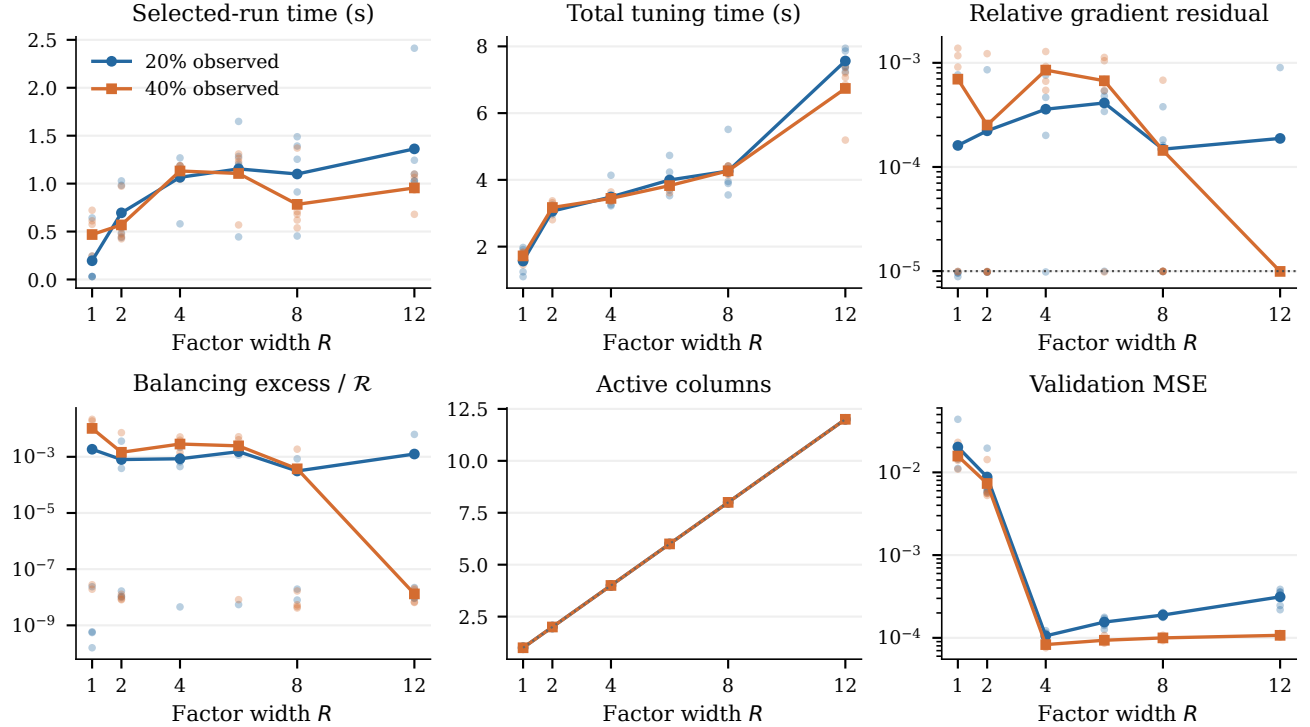


Figure 7: Additional width diagnostics. Curves are means and faint dots show individual results across five seeds. The gradient panel’s dotted line is the common stopping tolerance; the active-column panel’s dotted line is R . Selected-run and total tuning times are shown separately. Active columns are a factorization diagnostic, not an estimated CP rank.

A deterministic rank-one construction. For a unit-norm rank-one atom $\mathcal{U}$ and weight $\mu > 0$, splitting $\mu\mathcal{U}$ into k identical directions of equal weight μ/k and balancing every column yields

$$\frac{k\phi_N(\mu/k)}{\mu} = \cosh\left[\left(1 - \frac{2}{N}\right)\log\left(\frac{\mu}{k}\right)\right].$$

Figure 8 evaluates this construction for $N \in \{3, 4\}$, $\mu \in \{0.25, 1, 4, 16\}$, and $1 \leq k \leq 32$. The represented tensor and its atomic norm μ are unchanged. For integer μ , taking $k = \mu$ attains the atomic lower bound; further equal

splitting increases this construction’s cost. This does not contradict monotonicity of the globally optimized fixed-width objective, because the larger model can retain the earlier representation and pad with zero columns. The plotted equal-split costs are feasible constructions, not a claim to solve the general allocation problem for every μ and width.

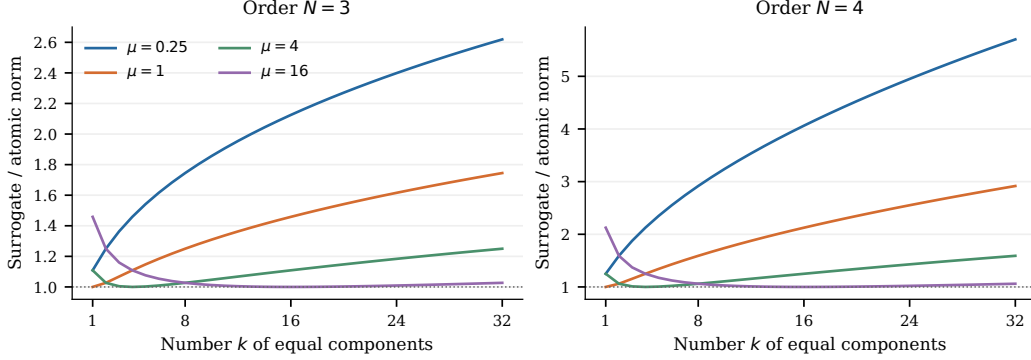


Figure 8: Effect of splitting one rank-one tensor into equal components. The tensor’s CP rank remains one. Each curve is the surrogate divided by the exact atomic norm for a specified feasible factorization, rather than the optimized width- R penalty. The dotted horizontal line is the atomic lower bound.

I ADDITIONAL COMPLETION EXPERIMENTS

Our evaluation covers four complementary aspects: the optimization benefit of exact cyclic block minimization, recovery accuracy on large real tensors, time–accuracy behavior under matched compute budgets, and runtime and memory scaling with tensor size, order, rank, and number of observations. We refer to the resulting exact cyclic solver as *Cyclic Block Optimization* (CBO).

I.1 Experimental Protocol

Data. Table 4 lists the five public tensors used in the main evaluation. Each tensor is independently min–max normalized to $[0, 1]$. The suite covers hyperspectral imaging, grayscale and color video, and two traffic systems. The fourth-order color video tests whether the implementation extends beyond order-three inputs.

Masks, noise, and metrics. For each tensor and seed, we sample an independent random-entry observation mask with either 20% or 10% observed entries, corresponding to 80% and 90% missingness. We use five seeds, shared by all methods. Measurement noise is added only to observed entries:

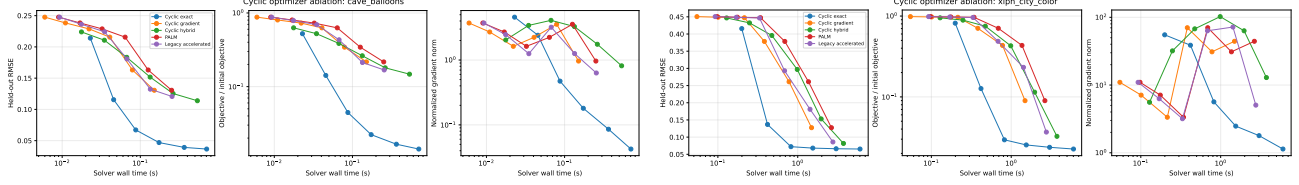
$$Y_\omega = X_\omega + \sigma(\max(X) - \min(X))\epsilon_\omega, \quad \epsilon_\omega \sim \mathcal{N}(0, 1), \quad (111)$$

where $\sigma \in \{0, 0.01, 0.05, 0.10\}$. Evaluation always uses the clean tensor on the missing entries $\mathcal{M} = \Omega^c$. The primary metric is

$$\text{RMSE}_{\mathcal{M}} = \sqrt{\frac{1}{|\mathcal{M}|} \|P_{\mathcal{M}}(\hat{\mathcal{X}} - \mathcal{X})\|_F^2}. \quad (112)$$

Baselines and applicability. We compare with PALM (Bolte et al., 2014) applied to the same factorized objective, CP-WOPT, HaLRTC (Liu et al., 2012), and LRTC-TNN (Zhang and Aeron, 2017). TT-WOPT (Yuan et al., 2019) is included for the fourth-order tensor, and LATC (Chen et al., 2022) is included only for traffic data. Mean filling is a non-iterative reference. The benchmark does not include CoSTCo, Bayesian CP, scaled-gradient or Riemannian Tucker methods, or TV Prox-Adan. Those methods are complementary; the results below compare only the listed baselines.

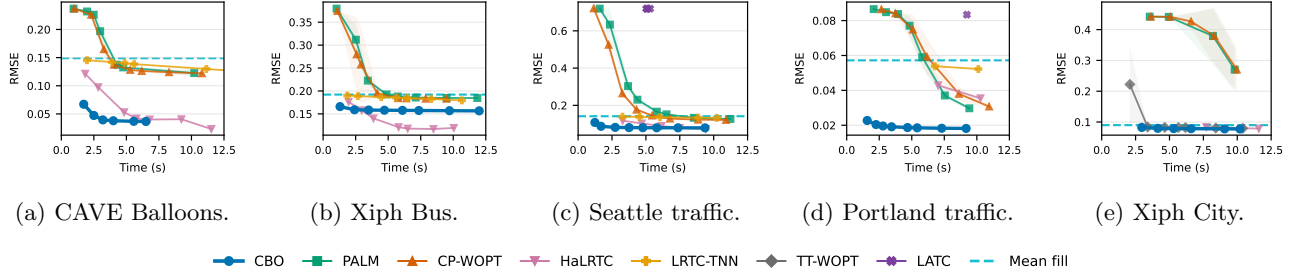
Hyperparameters and time matching. For each method–dataset pair, we reserve 10% of the observed entries for validation. Up to eight configurations are successively screened with 2-, 5-, and 10-second validation



(a) CAVE hyperspectral tensor.

(b) Fourth-order color video.

Figure 9: Held-out RMSE, normalized objective, and stationarity against solver wall time. Points are fresh deterministic runs at geometric sweep checkpoints; metric and stationarity computation are excluded from time.



(a) CAVE Balloons.

(b) Xiph Bus.

(c) Seattle traffic.

(d) Portland traffic.

(e) Xiph City.

—●— CBO —■— PALM —▲— CP-WOPT —▼— HaLRTC —◆— LRTC-TNN —◆— TT-WOPT —◆— LATC — — Mean fill

20% observed; three shared seeds; checkpoints calibrated at 1, 2, 3, 4, 5, 6, 8, and 10 s

Figure 10: Missing-entry RMSE versus actual solver wall time at 80% missingness with clean observations. complete-sweep checkpoints calibrated at nominal 1, 2, 3, 4, 5, 6, 8, and 10 seconds; their horizontal positions are actual median solver times. The dashed horizontal line is mean filling. Lower is better.

stages. Selection uses validation seed zero only; the selected configuration is then frozen and refit using all observed entries. CBO selects exact block minimization on all five tensors, with $(R, \lambda) = (8, 10^{-3})$ on CAVE and $(R, \lambda) = (4, 10^{-4})$ on the remaining tensors. The complete search grids and selected baseline configurations must be supplied with the reproducibility materials. The reported five-second solver-time target excludes this screening cost. Each iterative method is evaluated against this nominal time target; the achieved timing ratios are reported below. Data loading and metric computation are excluded from solver time. The experiment ran in Python 3.11 with NumPy 2.3 on a 10-core Apple M4 MacBook Air with 32GB RAM.

I.2 Extreme Missingness and Observation Noise

Table 5 gives the clean-observation results. At a five-second horizon, CBO has the lowest mean RMSE in nine of the ten dataset-missingness settings. The exception is Xiph Bus with 80% missing entries, where HaLRTC obtains 0.1402 RMSE compared with 0.1573 for CBO. At 90% missingness the ordering reverses on the same video. The largest margins occur on the traffic tensors; for example, on Portland with 80% missing entries, CBO obtains 0.0188 RMSE compared with 0.0507 for the strongest baseline.

Across all five datasets, two missingness levels, and four noise levels, CBO wins 36 of 40 dataset-condition cells and has mean rank 1.10. All four losses are the Xiph Bus conditions at 80% missingness. HaLRTC wins those four cells and has mean rank 2.10. TT-WOPT has mean rank 2.00 on its eight applicable fourth-order cells. The calibrated CBO checkpoint lies in the predeclared 0.8–1.25 timing window in all 40 conditions. At $\sigma = 0.10$, the median RMSE increase of CBO relative to clean observations is only 0.44%, although the mean is 6.87% because Portland is more noise-sensitive. The increase reaches 19.85% and 30.72% on Portland at 80% and 90% missingness, respectively; its absolute RMSE nevertheless remains lower than all baselines. Figure 5 shows the complete rank matrix.

The apparently small change from 80% to 90% missingness for several factorized methods should be interpreted as a wall-time result, not an iteration-matched result: fewer observations make each sweep cheaper, allowing more sweeps within five seconds.

Table 4: Real tensors in the main evaluation.

Tensor	Modality	Shape	Entries
CAVE Balloons	hyperspectral	256×256×31	2.03M
Xiph Bus	gray video	144×176×72	1.82M
Seattle	traffic	323×28×288	2.60M
Portland	traffic	1156×31×96	3.44M
Xiph City	color video	144×176×3×72	5.47M

Table 5: Missing-entry RMSE with clean observations. Values are mean $\pm$ standard deviation over five masks. “Other” is the strongest applicable iterative baseline in that row. All iterative methods receive the same five-second target.

Tensor	Missing	CBO	Other	Method
CAVE	80%	0.03737 $\pm$ 0.00055	0.05306 $\pm$ 0.00017	HaLRTC
CAVE	90%	0.03761 $\pm$ 0.00132	0.13482 $\pm$ 0.00028	HaLRTC
Bus	80%	0.15728 $\pm$ 0.00137	0.14022 $\pm$ 0.00019	HaLRTC
Bus	90%	0.15660 $\pm$ 0.00042	0.17958 $\pm$ 0.00025	HaLRTC
Seattle	80%	0.08013 $\pm$ 0.00138	0.10278 $\pm$ 0.00019	HaLRTC
Seattle	90%	0.07927 $\pm$ 0.00250	0.12717 $\pm$ 0.00009	HaLRTC
Portland	80%	0.01878 $\pm$ 0.00043	0.05069 $\pm$ 0.00003	HaLRTC
Portland	90%	0.01913 $\pm$ 0.00117	0.05360 $\pm$ 0.00002	HaLRTC
City	80%	0.08024 $\pm$ 0.00149	0.08209 $\pm$ 0.00048	TT-WOPT
City	90%	0.07895 $\pm$ 0.00093	0.08181 $\pm$ 0.00077	TT-WOPT

I.3 Cyclic Update Ablation

We isolate the optimizer using CAVE Balloons ($64 \times 64 \times 31$) and Xiph City ($64 \times 80 \times 3 \times 48$), both with 90% missing entries. Every variant receives the same observations, rank $R = 8$, $\lambda = 10^{-3}$, and initial factors. We compare exact cyclic updates, Lipschitz gradient cyclic updates, an alternating exact/gradient hybrid, PALM, and the previous accelerated update. Fresh deterministic fits are evaluated after 1, 2, 4, 8, 16, and 32 complete sweeps for each of five seeds.

The Previous accelerated row is the legacy cyclic Nesterov block-gradient update, included only as an optimizer comparator. Set $t_0 = 1$ and $t_{k+1} = (1 + \sqrt{1 + 4t_k^2})/2$. At sweep k , let $\beta_k = (t_k - 1)/t_{k+1}$ and form

$$\mathbf{Z}_i^{k+1} = \mathbf{A}_i^k + \beta_k(\mathbf{A}_i^k - \mathbf{A}_i^{k-1}),$$

with zero momentum at the first sweep. In cyclic order, update

$$\begin{aligned} \mathcal{Z}^{k,i} &:= (\mathbf{A}_1^{k+1}, \dots, \mathbf{A}_{i-1}^{k+1}, \mathbf{Z}_i^{k+1}, \mathbf{A}_{i+1}^k, \dots, \mathbf{A}_N^k), \\ \mathbf{A}_i^{k+1} &= \mathbf{Z}_i^{k+1} - \frac{1}{L_i^k} \nabla_i F(\mathcal{Z}^{k,i}). \end{aligned}$$

This baseline is not the proposed CBO update, and we make no monotonicity or convergence claim for it.

Exact updates cost more per sweep but reach substantially lower recovery error and lower stationarity measures by the final checkpoint. On CAVE, their normalized gradient norm is 4.24×10^{-2} , compared with 6.32×10^{-1} for the previous accelerated update and 9.64×10^{-1} for gradient cyclic/PALM. On City, exact cyclic reaches 1.14, compared with 5.05 for the previous update and 44.3 for gradient cyclic/PALM. Across all 300 fitted trajectories, no objective increase exceeds the numerical tolerance. For the exact, gradient, and hybrid cyclic updates, this observation is consistent with the descent theorem. The accelerated comparator is outside that theorem’s scope.

I.4 Accuracy and Time

The five-second comparison in Section I.2 summarizes one operating point. To examine how recovery evolves with additional computation, we reuse the clean-observation, 80%-missing trajectories from the tuned large-tensor experiment. Each method-dataset configuration is selected before testing and then frozen. We evaluate

Table 6: Optimizer ablation at 32 sweeps. Each cell reports missing-entry RMSE / median solver seconds; RMSE is mean over five shared masks.

Optimizer	CAVE	City
Exact cyclic	0.0366 /0.706	0.0654 /6.025
Gradient cyclic	0.1306/0.151	0.1274/1.490
Hybrid cyclic	0.1138/0.537	0.0816/3.739
PALM	0.1306/0.251	0.1274/2.641
Previous accelerated	0.1207/0.255	0.0859/2.755

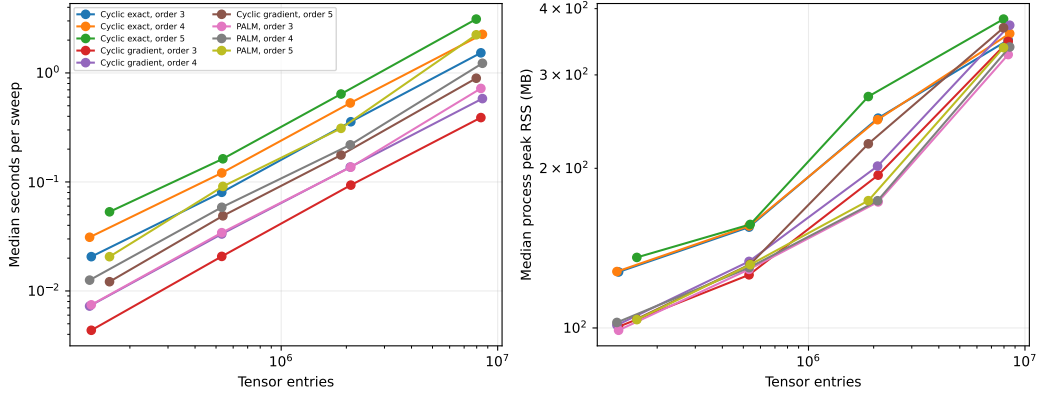


Figure 11: Median seconds per sweep and fresh-process peak resident memory versus tensor entries. Lines distinguish optimizer and tensor order. Both axes are logarithmic.

fresh deterministic fits on three shared masks at nominal 1-, 2-, 3-, 4-, 5-, 6-, 8-, and 10-second horizons. The horizontal coordinate in Figure 10 is the median measured solver time. If adjacent targets map to the same complete-sweep count, that fresh fit appears only once. Across the 38 dataset-budget cells containing at least one iterative method whose median solver time lies in the predeclared timing window, CBO obtains the lowest mean RMSE in 26. HaLRTC wins nine cells, while PALM, CP-WOPT, and TT-WOPT each win one. No method falls inside the timing window for the one-second Portland and Xiph City cells, which are excluded from this count.

I.5 Scalability

We generate sparse observations directly from CP factors, without materializing a dense tensor. We vary tensor order in $\{3, 4, 5\}$ and total entries from approximately 1.3×10^5 to 8.5×10^6 at rank 8 and 10% observation rate. Separate sweeps vary rank $R \in \{4, 8, 16, 32\}$ and observation rate $p \in \{0.05, 0.10, 0.20, 0.40\}$ on an order-three tensor with 2.10 million entries. Every case uses three seeds and three complete sweeps. Each method-case-seed combination runs in a fresh single-thread process so peak resident memory is not inherited from previous fits. Figure 11 shows nearly linear time growth in the number of tensor entries. The empirical time exponents for exact cyclic updates are 1.043, 1.031, and 1.049 for orders 3, 4, and 5, respectively. At the largest tested sizes, exact cyclic updates require 1.53, 2.26, and 3.12 seconds per sweep, with peak resident memory of 349, 359, and 382 MB. The corresponding time exponents are 1.047–1.094 for cyclic gradient and 1.084–1.182 for PALM. Rank is the principal cost driver for exact block solves. At 2.10 million entries and 10% observations, increasing R from 4 to 32 increases exact-cyclic time from 0.211 to 8.689 seconds per sweep and peak memory from 165 to 1162 MB. Increasing the observation rate from 5% to 40% at rank 8 increases time from 0.249 to 2.730 seconds per sweep. All 180 scaling fits are numerically monotone.

I.6 Synthetic Norm and Surrogate-Gap Diagnostics

We complement the recovery experiments with direct checks of the gap analysis. For 400 synthetic CP factorizations of order three or four, we sample dimensions in $\{2, \dots, 7\}$, factor widths in $\{1, \dots, 6\}$, and Gaussian factor entries with independent lognormal column scales. These deliberately imbalanced factors test the iden-

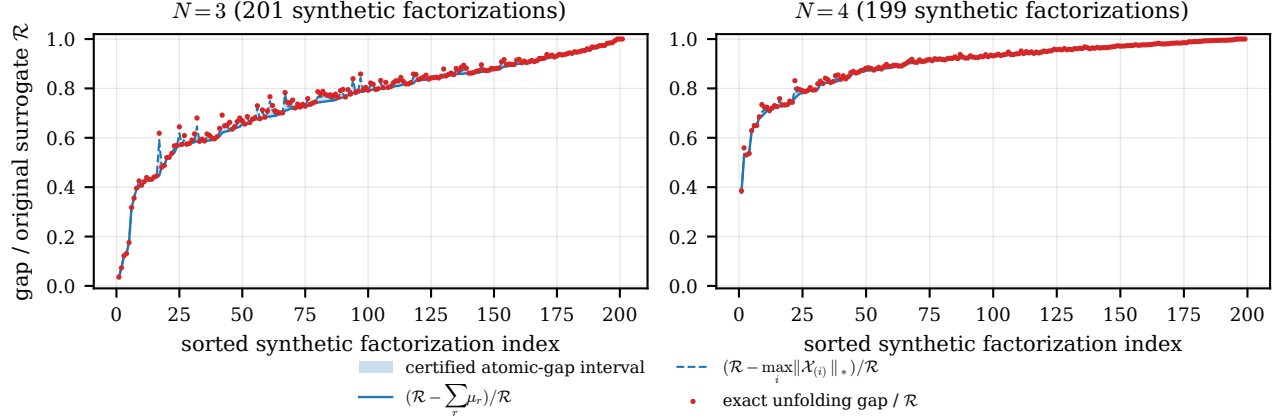


Figure 12: Surrogate-gap diagnostics for synthetic CP factors. Within each order, factorizations are sorted by their certified atomic-gap lower endpoint. The interval endpoints are given by Eq. (113); the unfolding gap is computed by SVD. All gaps are normalized by the regularizer of the original factorization. These generated factors illustrate the gap analysis and do not measure tightness at learned completion solutions.

tities across a range of gaps; they are not fitted completion solutions. Selected trials also contain zero-weight components to test the boundary cases. For each represented tensor, define $\rho := \mathcal{R}(\mathbf{A}_1, \dots, \mathbf{A}_N)$, $a := \sum_r \mu_r$, and $b := \max_i \|\mathbf{X}_{(i)}\|_*$. Then

$$1 - \frac{a}{\rho} \leq \frac{\rho - \|\mathbf{X}\|_{\text{at}}}{\rho} \leq 1 - \frac{b}{\rho}. \quad (113)$$

Figure 12 shows these certified intervals and the exact normalized unfolding gap $1 - \|\mathbf{X}\|_{\text{unf}}/\rho$.

The scaling identity has maximum error, normalized by ρ , of 8.46×10^{-16} . The Bures identity has errors at most 5.78×10^{-16} in the full-rank group and 2.35×10^{-9} in the singular group, classified using the numerical column ranks of the factors. Separate rank-one scaling experiments compare numerical minimization with $g_N(\mu)$; here the atomic norm is exactly μ . Lognormal Monte Carlo experiments evaluate the expected minimum scaling gap per component. Full protocols and numerical details are in Appendix J.

We also test independent standard Gaussian tensors at six sizes for each of orders three and four. All 200 unfolding-norm draws per size lie inside the stated probability interval. The fitted growth exponents are 2.008 and 2.506, respectively, close to the predicted 2 and 2.5. For the atomic norm, we use deterministic unfolding lower bounds and feasible slice-SVD upper bounds. The revised U_δ is 1.191–1.428 times the sample-mean slice-SVD certificate, and every certificate in the 30 draws per size lies below U_δ . These compare computable certificates with the theory; they do not compute the exact atomic norm of a general Gaussian tensor.

I.7 Scope of the Empirical Claims

The five-second table and the time–accuracy curves, which extend to ten seconds, measure finite-budget performance. We do not report a comparison at a common stationarity tolerance or after all methods have converged. The ranking reversal on Xiph Bus is visible in the reported means: at 80% missingness HaLRTC reaches 0.1402 RMSE versus 0.1573 for CBO, whereas at 90% missingness CBO reaches 0.1566 versus 0.1796 for HaLRTC. The masks at the two missingness levels are independently sampled, and a fixed time budget permits a different number of sweeps as the observation count changes. These results do not identify the cause of the reversal.

The optimizer ablation holds the objective and initial factors fixed, so it compares update rules. It does not isolate the effect of the surrogate regularizer from the solver: CP-WOPT uses a different objective and optimization procedure. Likewise, the validation-selected real-data ranks (8, 4, 4, 4, 4) are operating points, not a rank-sensitivity study; the real-data experiments do not vary factor initialization systematically. The tuning search is excluded from the five-second solver budget.

Finally, the synthetic experiments evaluate Eq. (47) and the Gaussian norm benchmarks. The controlled width



Figure 13: Top row: frames of colored video (4D) with massive missing observations; Bottom row: frames of video recovered.

study also evaluates fitted synthetic factors (Section 6.3 and Appendix H). These are separate from factors learned in the real-data completion benchmark. Numerical gap intervals for those fitted factors remain unreported, so the recovery experiments do not establish empirical tightness of the surrogate.

I.8 Qualitative Video Illustration

Fig. 13 illustrates observed frames and their corresponding reconstructions for color video.

J SYNTHETIC NORM AND SURROGATE-GAP EXPERIMENTS

We complement the completion benchmarks with synthetic checks of the deterministic gap identities and the Gaussian norm bounds. These experiments separate directly computable unfolding norms, certified atomic-norm bounds, and heuristic spectral estimates. The synthetic CP factorizations are generated independently of the completion algorithms; their gaps describe those factorizations rather than fitted factors from the completion benchmarks.

J.1 Protocol and reproducibility

All experiments use NumPy’s default random-number generator with seed 2027. For the Gaussian experiments, each tensor has independent $\mathcal{N}(0, 1)$ entries and equal mode sizes $I_i = n$. We use $n \in \{8, 12, 16, 24, 32, 48\}$ for $N = 3$ and $n \in \{4, 6, 8, 10, 13, 16\}$ for $N = 4$, with $\delta = 0.05$. At every order-size combination, we draw 200 tensors for the unfolding experiment and a separate set of 30 tensors for the atomic/spectral experiment. The nominal probability in Eqs. (55) and (65) is at least 0.90 for an individual tensor; these statements do not assert simultaneous coverage of every draw in an experiment.

For each tensor in the atomic/spectral experiment, the higher-order power method (HOPM) uses one initialization from the leading left singular vectors of the unfoldings and eight independent random unit-vector initializations. Each run performs at most 200 cyclic sweeps, stopping when the absolute change in its multilinear value is at most 10^{-10} times the larger of one and the current value. We retain the largest value over the nine starts. Matrix nuclear and operator norms are evaluated by SVD. The reproducibility package contains the corrected experiment script, configuration and run metadata, individual Gaussian measurements, summary CSV files, and the figures used here.

J.2 Gaussian unfolding-norm bounds

For each Gaussian draw, we directly evaluate $\|\mathcal{G}\|_{\text{unf}} = N^{-1} \sum_i \|G_{(i)}\|_*$ and compare it with the bounds in Proposition 8. Figure 14 shows the sample means and empirical 2.5–97.5% quantiles. These quantiles describe

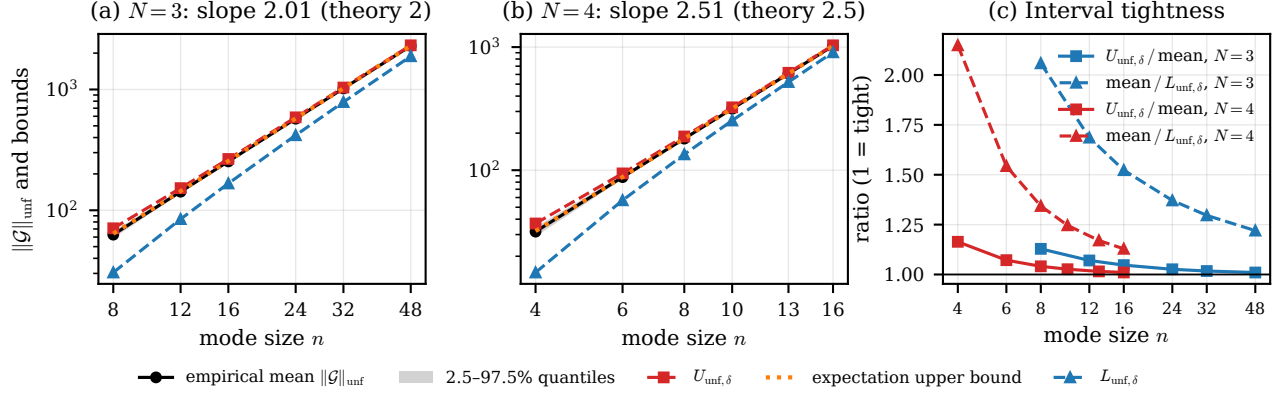


Figure 14: Gaussian unfolding-norm experiment with 200 draws per size, $\sigma = 1$, and $\delta = 0.05$. The norm is evaluated directly by matrix SVDs. Panels (a)–(b) compare its sample mean and empirical quantiles with the probability and expectation bounds; panel (c) shows tightness ratios. The fitted finite-size slopes are consistent with the dimension order established in Proposition 8.

variation between draws, not uncertainty in the sample mean. The least-squares slopes of $\log(\text{sample mean})$ against $\log n$ are 2.008 and 2.506 for $N = 3$ and $N = 4$, respectively, consistent with the predicted exponents 2 and $5/2$. Across the tested sizes, the ratio of the expectation upper bound in Eq. (66) to the sample mean ranges from 1.0005 to 1.0183. Table 7 reports the observed finite-sample coverage and interval-to-mean ratios. These computations assess the bounds in the tested regimes; the probability and expectation guarantees follow from the preceding proofs.

J.3 Certified atomic bounds and spectral diagnostics

For a tensor $\mathcal{G}$, let

$$B_{\text{low}} := \max_i \|G_{(i)}\|_*, \quad B_{\text{slice}} := \min_{a < b} \sum_{\alpha} \|G_{\alpha}^{(a,b)}\|_*,$$

where $G_{\alpha}^{(a,b)}$ is the matrix obtained by fixing all indices except those in modes a, b , and α ranges over the fixed indices. Every slice SVD, tensored with standard basis vectors in the remaining modes, gives an atomic decomposition of that slice embedded in the tensor. Consequently,

$$B_{\text{low}} \leq \|\mathcal{G}\|_{\text{at}} \leq B_{\text{slice}}.$$

The shaded region in Figure 15 is bounded by the sample means of these two certificates. It brackets the unknown sample mean of the atomic norm, and is not a confidence interval.

We compare these certificates with L_{δ} and the revised fiber upper bound U_{δ} in Eqs. (53)–(54). For reference, the earlier coordinate-decomposition bound is

$$U_{\text{coord},\delta} = \sigma \left(D\sqrt{2/\pi} + \sqrt{2D \log(1/\delta)} \right).$$

Across the tested sizes, $U_{\text{coord},\delta}/U_{\delta}$ ranges from 1.649 to 5.538. This comparison illustrates the improved dimension dependence of the fiber bound. The revised U_{δ} divided by the sample mean of B_{slice} ranges from 1.191 to 1.428; this compares two upper bounds and does not determine their tightness relative to the unknown atomic norm. In Table 7, an upper-certificate success means $B_{\text{slice}} \leq U_{\delta}$, and a lower-certificate success means $B_{\text{low}} \geq L_{\delta}$. Either condition is sufficient to establish the corresponding inequality for that draw. Failure of a certificate is inconclusive, so these success rates are not measurements of the unknown true atomic-norm coverage.

The HOPM value $\hat{s}$ is a feasible rank-one contraction and hence $\hat{s} \leq \|\mathcal{G}\|_{\text{sp}}$. Although duality gives Eq. (57), substituting this lower estimate in its denominator does not preserve an atomic lower bound. We retain $\|\mathcal{G}\|_F^2/\hat{s}$ in the CSV files only as a heuristic duality proxy; it is excluded from the plots and does not certify either side of the atomic interval. The spectral diagnostic displays the sample-average HOPM values against the expectation

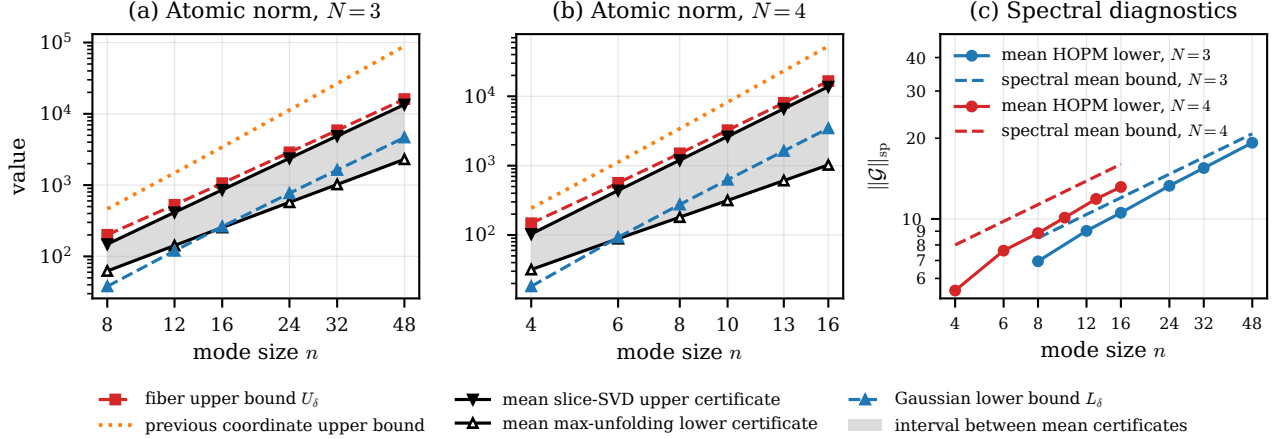


Figure 15: Atomic certificates and spectral diagnostics with 30 Gaussian draws per size. The shaded atomic interval uses averages of the certified unfolding lower bound and slice-SVD upper bound. The revised Gaussian fiber bound is compared with the coordinate bound. Panel (c) compares HOPM spectral lower estimates with the theoretical expectation upper bound.

upper bound $\sigma \sum_i \sqrt{I_i}$. Its being above the HOPM curve is a comparison with a lower estimate, rather than a measurement of the true spectral norm.

J.4 Deterministic identities and direct surrogate-gap diagnostics

We generate 400 CP factorizations. For each trial, we independently sample N uniformly from $\{3, 4\}$, each mode size uniformly from $\{2, \dots, 7\}$, and the CP width $\tilde{R}$ uniformly from $\{1, \dots, 6\}$. Each factor column is a standard Gaussian vector multiplied by an independent $\exp(Z)$, $Z \sim \mathcal{N}(0, 1)$, producing unequal modewise scales. To exercise zero-component cases, we set the first column of the first factor to zero when the zero-based trial index is divisible by 17, and the first column of the last factor to zero when it is divisible by 29. Products excluding a mode are evaluated directly, without division by a possibly zero factor norm. This yields 37 cases with a zero-weight component, including seven width-one cases representing the zero tensor. Every original regularizer is positive, so the normalization remains well defined; the normalized original gaps in those seven cases equal one.

We compare both sides of Eq. (33) and the Bures–Wasserstein identity in Eq. (60). The maximum absolute discrepancy divided by $\mathcal{R}$ is 8.46×10^{-16} for the scalar decomposition. For the Gram-matrix identity it is 5.78×10^{-16} in the 174 cases with all factors numerically full column rank, and 2.35×10^{-9} in the 226 remaining cases. Full column rank is equivalent to a nonsingular factor Gram in exact arithmetic; the numerical classification uses the ranks of the factors. Matrix square roots are computed by symmetric eigendecomposition, with negative eigenvalues arising from roundoff clipped to zero; the Bures cross term is evaluated as a nuclear norm using SVD. Singular matrices are more sensitive to floating-point evaluation of square roots. These finite numerical discrepancies are reported explicitly rather than interpreted as failures of the analytic identity.

Figure 3 summarizes these checks. For the same factorizations, Figure 12 reports the certified atomic-gap interval from Eq. (47) and the directly computed unfolding gap. The atomic interval does not require a norm-attaining CP representation. The accompanying CSV files also record diagnostics for the balanced representation: zero-weight components are removed and nonzero components are balanced according to Eq. (38); both representations produce the same tensor.

J.5 Optimal scaling and lognormal amplitudes

For $N \in \{3, 4, 5\}$ and 17 logarithmically spaced amplitudes $\mu \in [10^{-2}, 10^2]$, we numerically minimize the component surrogate under $\prod_i u_i = \mu$. We parameterize $\log u_1, \dots, \log u_{N-1}$ freely and set $\log u_N = \log \mu - \sum_{i < N} \log u_i$. An exact-derivative trust-region method uses five independent Gaussian starts with standard deviation 1.5 in

Table 7: Monte Carlo verification of the Gaussian bounds ($\sigma = 1$, $\delta = 0.05$; nominal coverage ≥ 0.90). Unfolding coverage uses directly computed norms. Atomic upper/lower certificate success rates are sufficient conditions, not the unknown true atomic coverage. The last column compares the previous coordinate bound with the revised fiber bound.

N	n	cov.	Unfolding		Atomic certificates and bound improvement		
			$\frac{U_{\text{unf},\delta}}{\text{mean}}$	$\frac{\text{mean}}{L_{\text{unf},\delta}}$	upper success	lower success	$U_{\text{old}}/U_{\delta}$
3	8	100%	1.13	2.06	100%	100%	2.31
3	12	100%	1.07	1.69	100%	100%	2.80
3	16	100%	1.05	1.52	100%	0%	3.22
3	24	100%	1.03	1.37	100%	0%	3.93
3	32	100%	1.02	1.30	100%	0%	4.53
3	48	100%	1.01	1.22	100%	0%	5.54
4	4	100%	1.16	2.15	100%	100%	1.65
4	6	100%	1.07	1.54	100%	0%	1.99
4	8	100%	1.04	1.34	100%	0%	2.28
4	10	100%	1.03	1.25	100%	0%	2.54
4	13	100%	1.02	1.17	100%	0%	2.89
4	16	100%	1.01	1.13	100%	0%	3.20

these coordinates, gradient tolerance 10^{-9} , and at most 1000 iterations. The objective is divided by the positive constant $\phi_N(\mu)$ for numerical scaling; its minimizers are unchanged. We retain the smallest finite objective and save optimizer statuses and residuals. The maximum discrepancy between the resulting relative gap and Eq. (40) is 3.55×10^{-15} . For a single rank-one tensor with coefficient μ and unit-norm mode vectors, the atomic norm is exactly μ . Thus this comparison also evaluates the exact optimized surrogate-atomic gap in that rank-one setting.

Finally, we draw 2×10^6 independent standard normal variables Z and form $M = \exp(\sigma_{\mu}Z)$, with $\eta = 0$ and $\sigma_{\mu} \in \{0, 0.1, \dots, 1\}$. For each $N \in \{3, 4, 5\}$, we compare the sample mean of $g_N(M)$ with Eq. (45) and the small-variance expansion in Eq. (46). The largest discrepancy from the exact expectation is 1.52 estimated Monte Carlo standard errors over the nondegenerate settings. The same Z samples are reused across orders and variances, so comparisons between settings are correlated. This experiment measures the expected minimum scaling gap per component. Its identification with the expected total atomic gap requires the norm-attaining representation assumption stated before Eq. (43).